

Large cardinals at the inconsistency frontier: a synthesis

Deduplicated results of the eighteen continuation reports on cover-exacting, exacting, and ultraexacting cardinals

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Purpose. Eighteen independent continuation reports were written in two rounds. The first nine (R1–R9, in docs/research-reports/01–09) continue the research plan, the *Unified Report*; seven of them prove essentially the same central theorem with different names, thresholds, and proof routes. The second nine (R10–R18, in 10–18) continue the first edition of this synthesis; all nine prove the same sharp-width theorem for definable sections of the cofinal quotient, again with different refinements. This document merges them. Each result is stated once, in the strongest form reached by any report, with a compact proof and a source tag such as [R1, R6]. Section 13 compares the reports and records the discrepancies found.

The six result groups

A. Completeness barrier (R1–R4, R6–R8). If λ is γ -cover exacting then λ is regular in $\text{HCD}(\gamma^+)$, and in $\text{HCD}(\gamma)$ when γ is singular, although $\text{cf}^V(\lambda) = \omega$. A strongly compact cardinal *below* λ forces a small, explicit witness to non-stabilization of the HCD hierarchy; a strongly compact *above* γ forces a proper ground and contradicts the Ground Axiom.

B. Exacting cardinals (R9; R10, R14, R17, R18). No OD_{V_λ} family of fewer than λ short cofinal subsets of λ exists; small definable families of subsets of V_λ are separated at a bounded rank; no definable countably additive law on cofinal sequences exists.

C. Ultraexacting cardinals (R5, R7). Definable finite choice fails on the quotient of cofinal ω -subsets of λ modulo finite difference: no finite OD_{V_λ} family of selectors, no ordinal-definable finite-valued transversal, and a forbidden Icarus enrichment.

D. Consistency (R1–R3, R5, R7–R9). $\text{Con}(\text{ZFC} + \text{I3}_{\text{wf}}(0))$ implies the consistency of a λ -cover-exacting λ ; and one I0-equiconsistent theory carries all the boundary phenomena of groups A–C simultaneously.

E. Class embeddings (R3). For $j : V \rightarrow M$ and a fixed regular θ , no unbounded subset of $j''\theta$ belongs to M .

F. Sections of the cofinal quotient (R10–R18). At an ultraexacting λ there are at least λ *rigid* classes q of cofinal ω -sets modulo finite difference: every set of representatives definable from q and parameters in V_λ meets q in 0 or λ points. Hence every definable complete section has λ fibres of full size λ , and the definable width of the quotient is exactly λ . The equivariant finite-label problem of group C is solved exactly: a definable rule exists iff every single permutation fixes a

label (R13). With a strongly compact cardinal below λ , an inner model in which λ is regular has its successor of λ collapsed or a canonical stationary set destroyed (R14).

Status. Nothing here is an unconditional inconsistency of a standard large-cardinal hypothesis; all eighteen reports agree on that. The open problem of a cover-exacting cardinal above a strongly compact cardinal remains open. The proofs are conventional, unrefereed, and not machine-checked. The synthesis re-derived the arguments while merging them, but did not re-verify the external sources; theorem numbers for imported results are those quoted by the reports (those used in the Lean formalization were later checked against the sources; see Cardinals/README.md). Priority is not claimed for any statement. *Second edition:* Sections 7, 8, 10, 13 and 14 were amended and Section 12 is new; the numbering of all first-edition statements is unchanged.

Contents

1	The source reports	4
2	Conventions and imported results	5
3	Core lemmas on exacting witnesses	7
4	From covers to small exact hulls	8
5	The completeness barrier	8
6	A strongly compact cardinal below λ	9
7	A strongly compact cardinal above γ: grounds	11
8	Exacting cardinals: definable families of cofinal maps	13
9	Ultraexacting cardinals: failure of definable finite choice	14
10	Consistency results	16
11	Class embeddings: the internal unbounded-range obstruction	18
12	Sections of the cofinal quotient	19
13	Comparison of the reports	23
14	What remains open	25
A	Concordance	26
B	Directory map	28
	References	28

1 The source reports

The original archive folders had colliding names (several were called `Large_Cardinals_Research_Continuation`, with browser-style suffixes). They were renamed to numbered directories in order of file modification time, and one redundant level of nesting was removed.

	Title (main file)	Distinctive content
R1	<i>Frontier II: Cofinality barriers for cover-exacting cardinals</i> (Large_Cardinals_Continuation)	Barrier for all $\eta > \gamma$; promotion principle LP; ground theorem with forcing-size bound; failed-stationarity certificate; I2 calibration.
R2	<i>Cover exactingness, definability, and conditional inconsistency</i> (Cover_Exacting_Definability_Followup)	No-small-cover form of the gap; weak covering principle Cov; first I0 boundary package.
R3	<i>Local barriers at the large-cardinal frontier</i> (Large_Cardinals_Research_Continuation)	Chain-condition form of the ground obstruction; internal unbounded-range theorem for $j : V \rightarrow M$; first-HOD-gap theory.
R4	<i>Cover-exacting cardinals and the Ground Axiom</i> (Large_Cardinals_Research_Continuation)	Barrier proved <i>without</i> the stationary characterization (uniform covers, envelopes, absorption); ultrafilter-code promotion failure.
R5	<i>Finite choice at the ultraexacting boundary</i> (Large_Cardinals_Finite_Choice_Continuation)	Finite-family selector obstruction with cycle amplification; selector-coded Icarus corollary; bounded/cofinal quotient separation; finite checks (Python).
R6	<i>Cover-exacting cardinals and high-completeness cores</i> (Cover_Exacting_High_Completeness_Cores)	Singular endpoint $q(\gamma)$; proof of $\gamma \geq \lambda$; cofinal-refinement principle; first-regularization theorem.
R7	<i>Local complete-definability barriers, finite symmetry, and an I0-calibrated consistency result</i> (Large_Cardinals_Continuation)	Finite-label symmetry theorem; finite-valued transversal theorem; small-ground formulation; finite checks (Python).
R8	<i>Cover-exacting cardinals and completely definable grounds</i> (Large_Cardinals_Continuation)	λ strongly inaccessible in the ground; rank-embedding witnesses escape $\text{CD}(\gamma^+)$; first-disagreement profile.
R9	<i>Frontier: a Prikry consistency refinement and local definability obstructions</i> (Large_Cardinals_Research_Continuation)	Prikry construction of a cover-exacting cardinal from $\text{I3}_{\text{wf}}(0)$; projection-width theorem for exacting cardinals.

The second round arrived as nine zip archives with overlapping internal folder names (three of them contained a folder `Cardinals3_Continuation` or `Cardinals_Continuation`). They were extracted into 10–18, again in order of source modification time and without the extra nesting level.

	Title (main file)	Distinctive content
R10	<i>Beyond finite choice</i> (Beyond_Finite_Choice)	Small invariant families at <i>exacting</i> cardinals; no conversion of classes into short cofinal information; probability laws and kernels; bounded-separation characterization.
R11	<i>Quotient parameters and sharp uniformization barriers</i> (Large_Cardinals_Quotient_Continuation)	The models N_q ; classes that do and do not reveal cofinality; λ recodings with the same N_q ; minimum-order-type compression.
R12	<i>Maximal-width uniformization</i> (Maximal_Width_Uniformization)	Roots and orbits of e ; coordinate width; integer phase spectrum; envelopes and edit budgets; two sharpness examples. Finite checks (Python).
R13	<i>Cyclic symmetry at the ultraexacting boundary</i> (Cyclic_Symmetry_Ultraexacting)	Exact finite-label classification via cyclic stabilizers of incidence words; the five-label rule; both output types occur λ times. Finite checks (Python).
R14	<i>Small definable sections and stationary-correctness barriers</i> (Cardinals_Continuation)	Small collections of sections; successor stationary-set obstruction; (λ^+) -chain-condition bound; localization of the destroyed stationary set.
R15	<i>Orbit saturation</i> (Orbit_Saturation)	Zero-or-full width for small families of <i>partial</i> sections; thin covers; choiceless inner models; classification and count of periodic classes; $ Q_\lambda = 2^\lambda$.
R16	<i>Small fibres at the ultraexacting boundary</i> (Small_Fibres_Ultraexacting)	Tail-invariant hulls; ordinal colourings; coarser equivalence relations; predicate-expanded I1 embeddings.
R17	<i>Thin sections and simultaneous saturation</i> (Thin_Sections_and_Saturation)	One set of λ classes works for <i>all</i> OD_{V_λ} sets at once; fixed bijection $\lambda \rightarrow V_\lambda$; sharpness of the parameter rank.
R18	<i>Cofinal-orbit rigidity</i> (Cofinal_Orbit_Rigidity)	Rigidity relative to the class q itself as a parameter; the minimal-rank-parameter device; homomorphisms into relations with small classes.

A result-by-report concordance is given in Appendix A.

2 Conventions and imported results

2.1 Conventions

We work in ZFC. Unadorned cardinalities, completeness, clubs and stationarity are computed in V ; $|x|^N$ and cf^N are computed in an inner model N . For an inner model $N \models \text{ZF}$, “ $\text{cf}^N(\lambda) = \lambda$ ” means that N contains no cofinal map from an ordinal below λ into λ ; no Choice in N is needed. All embeddings in Sections 3–9 are sets, and elementarity is expressed through satisfaction for set structures.

Definition 2.1 (Exacting, ultraexacting). A cardinal λ is *exacting*, $\text{Ex}(\lambda)$, if for every cardinal $\zeta > \lambda$ there are $X < V_\zeta$ with $V_\lambda \cup \{\lambda\} \subseteq X$ and an elementary $j : X \rightarrow V_\zeta$ with $j(\lambda) = \lambda$ and $j \upharpoonright \lambda \neq \text{id}$. It is *ultraexacting*, $\text{UEx}(\lambda)$, if moreover $j \upharpoonright V_\lambda \in X$ can be arranged. The domain X need not be transitive.

Definition 2.2 (Relative and cover exactingness). For $\alpha > \lambda$ and $Y \subseteq V_\alpha$, a *relative λ -exacting witness* is an elementary $j : (X, \in, X \cap Y) \rightarrow (V_\alpha, \in, Y)$ with $(X, \in, X \cap Y) < (V_\alpha, \in, Y)$, $V_\lambda \cup \{\lambda\} \subseteq X$, $\text{crit}(j) < \lambda$, $j(\lambda) = \lambda$. For a set Y , write $\text{REx}(\lambda, Y)$ if such witnesses exist at every $\alpha > \lambda$ with $Y \subseteq V_\alpha$. For infinite cardinals $\lambda \leq \gamma$, λ is *γ -cover exacting*, $\text{CEx}_\gamma(\lambda)$, if

$$\forall \alpha > \lambda \forall y \in V_\alpha \exists Y \subseteq V_\alpha (y \in Y \wedge |Y| \leq \gamma \wedge \text{there is a relative witness for } (V_\alpha, Y)).$$

The bound γ is fixed before α and y ; the condition is $y \in Y$, not $y \subseteq Y$; the witness need not fix y .

Some reports define REx only at all sufficiently large heights. By Lemma 3.3 the two conventions agree for every use below.

Definition 2.3 (Complete definability). Let $\text{UF}_\eta(\text{Ord})$ be the class of proper η -complete ultrafilters on ordinals (closed under intersections of fewer than η members). $\text{CD}(\eta)$ is the class of sets definable from finitely many ordinals and finitely many members of $\text{UF}_\eta(\text{Ord})$, and $\text{HCD}(\eta) = \{x : \text{tc}(\{x\}) \subseteq \text{CD}(\eta)\}$; $\text{HCD} = \bigcap_\eta \text{HCD}(\eta)$. For $\eta \leq \xi$,

$$\text{CD}(\xi) \subseteq \text{CD}(\eta), \quad \text{HCD}(\xi) \subseteq \text{HCD}(\eta), \quad (2.1)$$

and for a set of ordinals a : $a \in \text{CD}(\eta) \iff a \in \text{HCD}(\eta)$.

Put $q(\gamma) = \gamma$ if γ is singular and $q(\gamma) = \gamma^+$ otherwise. A *ground* is an inner model $W \models \text{ZFC}$ with $V = W[G]$ for a set forcing $\mathbb{P} \in W$; the *Ground Axiom* GA says there is no proper ground.

2.2 Imported results

Input A (Local Kunen). In ZFC there is no nontrivial elementary $V_{\mu+2} \rightarrow V_{\mu+2}$. [7, 8]

Input B (Goldberg's HCD theorems). $\text{HCD}(\eta) \models \text{ZF}$ for every infinite cardinal η . If δ is strongly compact, then $\text{HCD}(\delta) \models \text{ZFC}$, it is a ground of V by a forcing of size below $(2^{2^\delta})^+$, and it has the δ -cover and δ -approximation properties in V . If δ is extendible, $\text{HCD}(\delta) = \text{HCD}$. [3, Prop. 4.2–4.3, Thm. 4.4, 4.10, Prop. 4.11, Thm. 4.14–4.15]

Several reports stress that the ground and covering statements need only *strong compactness*; the supercompact hypothesis in the shorter lecture-note summary is not needed. R1 notes that the arXiv v2 numbering is shifted by one relative to the author version used here.

Input C (Blue–Goldberg cover-exacting notes). (i) $\text{CEx}_\gamma(\lambda)$ implies $\gamma \geq \lambda$. (ii) Lemma 3.3: $\text{REx}(\lambda, Y)$ and $Z \in \text{OD}_{\{Y\}}$ imply $\text{REx}(\lambda, Z)$. (iii) Proposition 3.4: $\text{CEx}_\gamma(\lambda)$ iff every set belongs to some Y with $|Y| \leq \gamma$ and $\text{REx}(\lambda, Y)$, iff $\{\sigma \in P_{\gamma^+}(V_\alpha) : \text{REx}(\lambda, \sigma)\}$ is stationary for all large α . (iv) Theorem 3.5: $\text{Con}(\text{ZFC} + \text{I2}) \Rightarrow \text{Con}(\text{ZFC} + \exists \lambda \text{CEX}_\lambda(\lambda))$. (v) Theorem 3.6: no γ -cover-exacting cardinal lies above an extendible. (vi) Problem 5.2: can one lie above a strongly compact? [2]

Items (i)–(iii) are lecture-note results; R4 and R6 reprove everything needed from them (Section 4), so group A depends on the notes only through the definitions.

Input D (Exacting and ultraexacting cardinals). (i) An exacting λ is a strong limit of cofinality ω and is regular in HOD_{V_λ} [4, Thm. 2.10]. (ii) Ultraexacting witnesses can be chosen with critical point above any $\alpha < \lambda$ [4, Lemma 3.2]. (iii) λ is ultraexacting iff every ordinal-definable $A \subseteq V_{\lambda+1}$ admits an elementary $e : (V_{\lambda+1}, A) \rightarrow (V_{\lambda+1}, A)$ with $\text{crit}(e) < \lambda$ [5, Thm. B/3.1]. (iv) $\text{ZFC} + \exists \lambda \text{UEX}(\lambda)$ and $\text{ZFC} + \text{I0}$ are equiconsistent [5, Thm. A/4.5]. (v) An ultraexacting λ is preserved by a χ -directed closed forcing (any $\chi > \lambda$) forcing $V_\lambda \subseteq \text{HOD}$; then no exacting cardinal lies below λ [5, Prop. 3.9, Cor. 3.10].

Input E (Iteration interface for $\text{I3}_{\text{wf}}(0)$). For $j : V_\Lambda \rightarrow V_\Lambda$ with critical sequence $\langle \kappa_n \rangle$, finite iterates $j_{m,n}$, direct limit M_ω^j with well-founded part W_ω^j : j is an $\text{I3}_{\text{wf}}(0)$ -embedding if $j_{0,\omega}[\kappa_0^+] \subseteq W_\omega^j$. Then $V_\Lambda \cup \{\Lambda\} \subseteq W_\omega^j$; $j^+(j_{0,\omega}(A)) = j_{0,\omega}(A)$ for $A \in V_{\kappa_0+1}$; if $j^+(E) = E \subseteq \Lambda^2$ then $j \upharpoonright \Lambda$ is elementary on (Λ, E) ; and $A \in j_{0,\omega}(U_0)$ iff A contains a tail of $\langle \kappa_n \rangle$, for $A \in \mathcal{P}(\Lambda) \cap W_\omega^j$. Moreover I2 is strictly stronger than $\text{I3}_{\text{wf}}(0)$, and $\exists \lambda \text{Ex}(\lambda)$ is strictly stronger than I3, in consistency strength. [6, §3], [5, Prop. 5.1, Def. 5.2, Thm. 5.4, 5.10, Cor. 5.11]

3 Core lemmas on exacting witnesses

Lemma 3.1 (Critical-sequence normalization). [R1–R8] Let λ be an uncountable cardinal and let e be a nontrivial elementary self-embedding of V_λ , or of $V_{\lambda+1}$ with $\text{crit}(e) < \lambda$. With $\kappa_n = e^n(\text{crit}(e))$ we have $\sup_n \kappa_n = \lambda$, and $e(\xi) > \xi$ for every $\xi \in [\kappa_0, \lambda)$. In particular every exacting witness j restricts to such an $e = j \upharpoonright V_\lambda$, so $\text{cf}^V(\lambda) = \omega$ and every ordinal below λ fixed by j lies below $\text{crit}(j)$.

Proof. If $\mu = \sup_n \kappa_n < \lambda$, the sequence lies in V_λ , $e(\mu) = \mu$, and $e \upharpoonright V_{\mu+2}$ contradicts Input A. For $\kappa_n \leq \xi < \kappa_{n+1}$, $e(\xi) \geq \kappa_{n+1} > \xi$. The restriction $j \upharpoonright V_\lambda$ is elementary because V_λ is definable from the fixed parameter λ and $V_\lambda \subseteq X$; transitivity of X is not used. $\square$

Lemma 3.2 (No preserved short cofinal set). [R1, R2, R4, R6, R7] If $a \subseteq \lambda$ is cofinal with $\text{ot}(a) < \lambda$, then $\text{REx}(\lambda, a)$ fails.

Proof. In $(V_\alpha, \in, a)$ the set a is definable from its predicate, so $a \in X$ and $j(a) = a$; hence $\tau = \text{ot}(a)$ and the increasing enumeration $h : \tau \rightarrow a$ are in X and fixed. By Lemma 3.1, $\tau < \text{crit}(j)$, so $j(h(\xi)) = j(h)(j(\xi)) = h(\xi)$ for all $\xi < \tau$. Thus the cofinal set a consists of fixed ordinals, contradicting Lemma 3.1. $\square$

Lemma 3.3 (Predicate bookkeeping). [R4, R6] Let j be a relative witness for (V_θ, Y) with $Y \in V_\theta$. (a) $Y \in X$ and $j(Y) = Y$; members of Y need not be fixed. (b) If $\text{cf}(\theta) > \omega$, there are relative witnesses for (V_β, Y) at every $\beta < \theta$ with $\beta > \lambda$ and $Y \subseteq V_\beta$.

Proof. (a) Y is the unique z with $\forall u(u \in z \leftrightarrow Y(u))$. (b) Existence of a witness at a height $\beta < \theta$ is computed correctly in V_θ . The least failing β is definable from λ and Y , hence lies in X and is fixed; restricting j to $X \cap V_\beta$ gives a witness at β . $\square$

Lemma 3.4 (Definability transfer). [R4, R6; focused form R1–R3, R8] If $\text{REx}(\lambda, Y)$ and $Z \in \text{OD}_{\{Y\}}$, then $\text{REx}(\lambda, Z)$. In particular no cofinal subset of λ of order type below λ belongs to $\text{OD}_{\{Y\}}$, and λ is regular in $\text{HOD}_{\{Y\}}$.

Proof. Let Z be the least counterexample in the canonical well-order of $\text{OD}_{\{Y\}}$ and β its least failing height; both are definable from λ and Y without further ordinal parameters. Take a witness for Y at a height θ reflecting these definitions. Then $Z, \beta \in X$ are fixed, the predicate $u \in Z$ is a definitional expansion, and restriction to V_β gives the excluded witness. The second assertion follows from Lemma 3.2. $\square$

The parameter issue, as every report records it

An arbitrary ordinal parameter in a definition need not be fixed by an exacting embedding. All proofs therefore first pass to a canonical least object definable from the fixed pair (Y, λ) and only then choose a witness at a reflecting height. The same device reappears in Lemma 9.3 for ultraexacting cardinals.

Lemma 3.5 (The cover bound). [R6; known from the notes] $\text{CEx}_Y(\lambda)$ implies $\gamma \geq \lambda$.

Proof. If $\gamma < \lambda$, take a cofinal $c : \omega \rightarrow \lambda$, a height α with $\text{cf}(\alpha) > \gamma$, and a cover $Y \ni c$ with witness j . Then $Y \in V_\alpha$ is fixed, $v = |Y| < \text{crit}(j)$, and a bijection $f : v \rightarrow Y$ in X shows $Y \subseteq X$ and $j''Y = Y$. Let $c_n \in Y$ be the n -fold preimage of c ; then $c(k) = e^n(c_n(k))$ with $e = j \upharpoonright V_\lambda$. Choosing $c(k) \geq \text{crit}(j)$ and n with $e^n(\text{crit}(j)) > c(k)$ gives a contradiction, since e^n maps ordinals below $\text{crit}(j)$ below $\text{crit}(j)$ and the others to at least $e^n(\text{crit}(j))$. $\square$

4 From covers to small exact hulls

The reports use two routes to obtain a small elementary substructure $\sigma < V_\alpha$ that is simultaneously (i) of size at most γ , (ii) a container for prescribed parameters, and (iii) a predicate with $\text{REx}(\lambda, \sigma)$.

Route S (R1–R3, R7, R8) imports the stationary characterization, Input C(iii), and intersects the stationary set with the club of elementary hulls containing the parameters.

Route H (R4, R6) proves the required hull directly from Definition 2.2:

Lemma 4.1 (Persistent small covers). [R4, R6] $\text{CEx}_\gamma(\lambda)$ holds iff every set y belongs to a set Y with $|Y| \leq \gamma$ and $\text{REx}(\lambda, Y)$.

Proof. ($\Rightarrow$) If y has no such cover, let $b(Y)$ be the least failing height of each candidate Y , and let $F(\eta)$ bound $b(Y) + \omega$ over candidates $Y \subseteq V_\eta$; this uses Replacement on sets only. A continuous recursion of length γ^+ gives θ with $\text{cf}(\theta) = \gamma^+$ closed under F . Cover exactingness at θ yields $Y \ni y$; since $|Y| < \text{cf}(\theta)$, $Y \in V_\beta$ for some $\beta < \theta$, so $b(Y) < \theta$, and Lemma 3.3(b) gives a witness at $b(Y)$. ($\Leftarrow$) Use $Z = Y \cap V_\alpha \in \text{OD}_{\{Y\}}$ and Lemma 3.4. $\square$

Lemma 4.2 (Exact hulls). [R4, R6] Assume $\text{CEx}_\gamma(\lambda)$. For every sufficiently large limit α and finite $A \subseteq V_\alpha$ there is σ with $A \subseteq \sigma < V_\alpha$, $|\sigma| \leq \gamma$, and $\text{REx}(\lambda, \sigma)$.

Proof. Code a Skolem system for V_α (with constants for A) as one function f , cover f by a persistent Y (Lemma 4.1), and let σ be the closure of ω under *all* functions $V_\alpha^{<\omega} \rightarrow V_\alpha$ that belong to Y . Then $|\sigma| \leq \gamma$, $\sigma < V_\alpha$, and σ is definable from Y and α alone, so Lemma 3.4 applies. The individually chosen f is not a parameter of the definition. $\square$

Lemma 4.3 (Trace reconstruction). [R1–R4, R6–R8] Let U be an η -complete ultrafilter on an ordinal τ , and let $U, \tau \in \sigma < V_\alpha$ with $|\sigma| < \eta$ and α large. Then $U \in \text{OD}_{\{\sigma\}}$. Hence for $x \in \text{CD}(\eta)$ with η regular uncountable, there is a club $C \subseteq P_\eta(V_\alpha)$ with $x \in \text{OD}_{\{\sigma\}}$ for all $\sigma \in C$.

Proof. Let $\beta = \min \bigcap (U \cap \sigma)$, which exists by completeness. For $A \in \sigma \cap \mathcal{P}(\tau)$ we get $A \in U \iff \beta \in A$, using that $\tau \setminus A \in \sigma$. U is the *unique* ultrafilter $W \in \sigma$ on τ with this trace: two such candidates would be separated by a set in σ , by elementarity. The definition uses only σ and the ordinals τ, β, α . The seed β need not belong to σ , U need not be principal, and uniqueness is among candidates *inside* σ only. $\square$

The mechanism is the one in the proof of Theorem 3.7 of the Blue–Goldberg notes; the reports' addition is the exact inequality $|\sigma| \leq \gamma < \eta$.

Lemma 4.4 (The singular endpoint). [R6] If γ is singular, every γ -complete filter is γ^+ -complete; hence $\text{CD}(\gamma) = \text{CD}(\gamma^+)$ and $\text{HCD}(\gamma) = \text{HCD}(\gamma^+)$.

Proof. Split an index set of size γ into $\text{cf}(\gamma)$ pieces of size below γ and intersect in two stages. $\square$

5 The completeness barrier

Theorem 5.1 (Completeness barrier). [R1–R4, R6–R8] Assume $\text{CEx}_\gamma(\lambda)$ and put $\rho = q(\gamma)$.

1. *Absorption.* $\text{REx}(\lambda, x)$ for every $x \in \text{CD}(\rho)$. [R4, R6]
2. No cofinal $a \subseteq \lambda$ with $\text{ot}(a) < \lambda$ (equivalently $|a| < \lambda$) belongs to $\text{CD}(\rho)$.

3. $\text{cf}^V(\lambda) = \omega$, while $\text{cf}^{\text{HCD}(\eta)}(\lambda) = \lambda$ for every cardinal $\eta \geq \rho$.
4. Every cofinal $b \subseteq \lambda$ in $\text{HCD}(\rho)$, or in any inner model contained in it, has $\text{ot}(b) = \lambda = |b|^V$. Thus a countable cofinal $c \subseteq \lambda$ satisfies $\min\{|a|^V : a \in \text{HCD}(\rho), c \subseteq a \subseteq \lambda\} = \lambda$. [R2, R4, R6]

In particular $\text{CEx}_\lambda(\lambda)$ implies that λ is regular in $\text{HCD}(\lambda)$. [R6]

Proof. By Lemma 4.4 we may take $\rho = \gamma^+$. (1) Let x be definable from γ^+ -complete $U_0, \dots, U_{m-1}$ and ordinals. By Lemma 4.2 (Route H), or by stationarity (Route S), choose $\sigma < V_\alpha$ containing the U_i with $|\sigma| \leq \gamma$ and $\text{REx}(\lambda, \sigma)$. Lemma 4.3 gives $x \in \text{OD}_{\{\sigma\}}$, and Lemma 3.4 gives $\text{REx}(\lambda, x)$. (2) follows from (1) and Lemma 3.2. (3) λ remains a cardinal in $N = \text{HCD}(\eta)$; a cofinal $f : \mu \rightarrow \lambda$ in N with $\mu < \lambda$ has range of V -cardinality below λ , hence of order type below λ , and this range lies in $\text{CD}(\rho)$ by (2.1). (4) The order type of b is at most λ and cannot be smaller. $\square$

Remark 5.2 (Threshold). R1, R3 and R8 state the barrier for all $\eta > \gamma$, R2, R4 and R7 at γ^+ , and R6 at $q(\gamma)$; by (2.1) and Lemma 4.4 the last is the strongest. The successor cannot be dropped for regular γ by this proof: σ may have size exactly γ , and a uniform γ -complete ultrafilter on γ contains all $\gamma \setminus \{\xi\}$, whose intersection is empty R1. No report claims the threshold is optimal.

Corollary 5.3 (What escapes the high-completeness models). [R3, R8] Assume $\text{CEx}_\gamma(\lambda)$ and $\eta \geq q(\gamma)$. No strictly increasing cofinal $c : \omega \rightarrow \lambda$ is ordinal definable from an η -complete ultrafilter on an ordinal, however large its underlying ordinal. No nontrivial elementary $e : V_\lambda \rightarrow V_\lambda$ belongs to $\text{CD}(\eta)$.

Proof. $\text{ran}(c)$ is a short cofinal set. From e one defines its critical sequence, which is cofinal by Lemma 3.1. $\square$

Corollary 5.4 (Certificate of failed cover exactingness). [R1] If $\lambda \leq \gamma < \eta$ and some cofinal $a \subseteq \lambda$ of size below λ lies in $\text{CD}(\eta)$, then for all large α a club of elementary hulls in $P_{\gamma^+}(V_\alpha)$ is disjoint from $\{\sigma : \text{REx}(\lambda, \sigma)\}$; so $\text{CEx}_\gamma(\lambda)$ fails.

Corollary 5.5 (Lower covering and approximation fail). [R1–R4, R6, R8] Assume $\text{CEx}_\gamma(\lambda)$, let $W \models \text{ZF}$ be an inner model with $W \subseteq \text{HCD}(q(\gamma))$, and let $c \subseteq \lambda$ be cofinal of order type ω . Then $c \notin W$, but $c \cap b$ is finite and belongs to W whenever $b \in W$ is well-orderable in W with $|b|^W < \lambda$. Consequently (W, V) fails the θ -cover and θ -approximation size tests for every uncountable cardinal $\theta \leq \lambda$.

Proof. λ is regular in W , so $b \cap \lambda$ is bounded, and a cofinal ω -sequence has finitely many terms below a bound. $\square$

This is compatible with Input B: when $W = \text{HCD}(\kappa)$ for a strongly compact $\kappa > \gamma$, approximation and covering hold at κ , where the test set $b = \lambda$ is allowed and $c \cap \lambda = c \notin W$.

6 A strongly compact cardinal below λ

Theorem 6.1 (Small-witness separation). [R1–R4, R6–R8] Suppose $\delta < \lambda \leq \gamma$, $\text{SC}(\delta)$ and $\text{CEx}_\gamma(\lambda)$; let $\rho = q(\gamma)$. There is a cofinal $a \subseteq \lambda$ with

$$a \in \text{HCD}(\delta), \quad |a|^{\text{HCD}(\delta)} < \delta, \quad \text{ot}(a) < \delta, \quad a \notin \bigcup_{\eta \geq \rho} \text{CD}(\eta).$$

Hence $\text{cf}^{\text{HCD}(\delta)}(\lambda) < \delta < \lambda = \text{cf}^{\text{HCD}(\eta)}(\lambda)$ for all $\eta \geq \rho$, and $\mathcal{P}(\lambda) \cap \text{HCD}(\rho) \subsetneq \mathcal{P}(\lambda) \cap \text{HCD}(\delta)$. Moreover:

1. every $b \in \text{HCD}(\rho)$ with $a \subseteq b$ has $|b|^V \geq \lambda$; [R2]
2. no cofinal $b \subseteq a$ belongs to $\text{HCD}(\rho)$; [R6]
3. some δ -complete ultrafilter U on an ordinal satisfies $U \notin \text{CD}(\rho)$. [R4]

Proof. Cover a countable cofinal $c \subseteq \lambda$ by $b_0 \in \text{HCD}(\delta)$ of internal size below δ (Input B) and let $a = b_0 \cap \lambda$. Its order type is below the cardinal δ and its enumeration lies in $\text{HCD}(\delta)$. Theorem 5.1(2),(4) give the rest; for (3), take U with $a \in \text{OD}_{\{U\}}$ and compose definitions. $\square$

The witness a need not be countable, even in V : strong compactness supplies a small *cover* of an ω -sequence, not the sequence. The forcing-size bound of Input B gives an independent proof of the weaker conclusion $\text{cf}^{\text{HCD}(\delta)}(\lambda) < \lambda$, since a forcing of size below the strong limit λ cannot singularize a regular λ R1.

6.1 The conditional inconsistency, in all its reported forms

Each report names a slightly different additional principle. For $\delta < \lambda \leq \gamma$ and $\rho = q(\gamma)$:

Principle	Source
P_1 $\mathcal{P}(\lambda) \cap \text{HCD}(\delta) = \mathcal{P}(\lambda) \cap \text{HCD}(\rho)$ (local stabilization)	R3, R4, R6, R8
P_2 $[\lambda]^{<\delta} \cap \text{HCD}(\delta) \subseteq \text{HCD}(\rho)$, sizes ambient (local promotion LP)	R1, R2, R3
P_3 as P_2 with size computed in $\text{HCD}(\delta)$ (LS)	R7
P_4 every $x \in [\lambda]^{<\delta} \cap \text{HCD}(\delta)$ is covered by some $b \in \text{HCD}(\rho)$ with $ b ^V < \lambda$ (Cov)	R2
P_5 every cofinal $x \in \text{HCD}(\delta)$ with $\text{ot}(x) < \delta$ has a cofinal subset in $\text{HCD}(\rho)$ (CR)	R6
P_6 every δ -complete ultrafilter on an ordinal lies in $\text{CD}(\rho)$	R4
P_7 some cofinal $x \subseteq \lambda$ with $ x < \lambda$ lies in $\text{CD}(\rho)$	R1, R4, R7

Corollary 6.2 (Conditional inconsistency below λ). [R1–R4, R6–R8] ZFC refutes $\text{SC}(\delta) \wedge \delta < \lambda \leq \gamma \wedge \text{CEx}_\gamma(\lambda) \wedge P_i$ for each $i \leq 6$, and refutes $\text{CEx}_\gamma(\lambda) \wedge P_7$ outright. The implications are $P_6 \Rightarrow P_1 \Rightarrow P_2 \Leftrightarrow P_3$, $P_2 \Rightarrow P_4$, $P_2 \Rightarrow P_5$, and under the hypotheses each of P_4, P_5 yields P_7 .

Proof. Apply each principle to the set a of Theorem 6.1. For $P_2 \Leftrightarrow P_3$: a subset of λ in $\text{HCD}(\delta)$ of ambient size below δ has order type below δ , so its enumeration in $\text{HCD}(\delta) \models \text{ZFC}$ makes it internally small; the converse is trivial. $P_6 \Rightarrow P_1$ holds for sets of ordinals by composing definitions. $\square$

None of the principles is derived from strong compactness. The equivalence $P_2 \Leftrightarrow P_3$ is an observation of this synthesis; it shows that the reports' differing size conventions do not produce different theorems.

Corollary 6.3 (The extendible boundary recovered). [R1–R4, R6–R8] If $\delta < \lambda$ is extendible, then $\neg \text{CEx}_\gamma(\lambda)$ for every γ .

Proof. $\text{HCD}(\delta) = \text{HCD} \subseteq \text{HCD}(\rho) \subseteq \text{HCD}(\delta)$, so P_1 holds; extendibles are strongly compact. For $\gamma < \lambda$ use Lemma 3.5. $\square$

This is Blue–Goldberg's Theorem 3.6, not a new result. The factorization shows what each hypothesis contributes: strong compactness supplies the short cofinal set in the low model; extendibility supplies the promotion to high completeness.

Theorem 6.4 (First regularization). [R6] For an ambient cardinal λ let $r(\lambda)$ be the least infinite cardinal η with $\text{cf}^{\text{HCD}(\eta)}(\lambda) = \lambda$, if one exists. The map $\eta \mapsto \text{cf}^{\text{HCD}(\eta)}(\lambda)$ is nondecreasing. If $\text{CEx}_\gamma(\lambda)$ then $r(\lambda) \leq q(\gamma)$; if also $\text{SC}(\delta)$ for some $\delta < \lambda$ then $r(\lambda) > \delta$. If $\text{CEx}_\lambda(\lambda)$ and λ is a limit of strongly compact cardinals, then $r(\lambda) = \lambda$ exactly.

The consistency of the last hypothesis is not asserted. The theorem does not show $\text{HCD}(\lambda) = \bigcap_{\eta < \lambda} \text{HCD}(\eta)$, and no single short cofinal set need lie in all the lower models.

7 A strongly compact cardinal above γ : grounds

Lemma 7.1 (Singularizing forcing is not λ -cc). [R3, R4, R6, R8; size form R1, R2, R7] Let $W \models \text{ZFC}$ be an inner model in which λ is regular uncountable, and let $V = W[G]$ for $\mathbb{P} \in W$ with $\text{cf}^V(\lambda) = \omega$ and λ a cardinal of V . Then $\mathbb{P}$ is not λ -cc in W ; it has an antichain of W -size λ , no dense subset of W -size below λ , and $|\mathbb{P}|^V \geq \lambda$. Without the regularity assumption: if $|\mathbb{P}|^W \leq \nu < \lambda$ for an infinite ν , then W contains a cofinal $a \subseteq \lambda$ with $|a|^W \leq \nu$.

Proof. If $\mathbb{P}$ were λ -cc, the sets B_n of values of a name $\dot{c}(n)$ decided by maximal antichains have W -size below λ , so $\bigcup_n B_n$ is bounded in λ by regularity in W , contradicting forced cofinality. For the last part take all values forced by some $q \leq p$. $\square$

Theorem 7.2 (Proper canonical ground). [R1–R4, R6, R8] Suppose $\lambda \leq \gamma < \kappa$, $\text{CEx}_\gamma(\lambda)$ and $\text{SC}(\kappa)$. Then $W = \text{HCD}(\kappa)$ is a *proper* ground of V with

$$W \models “\lambda \text{ is strongly inaccessible}”, \quad \text{cf}^V(\lambda) = \omega.$$

Every $\mathbb{P} \in W$ with $V = W[G]$ satisfies the conclusions of Lemma 7.1, and (W, V) fails θ -covering and θ -approximation for every uncountable $\theta \leq \lambda$ while satisfying both at κ .

Proof. $\kappa \geq \gamma^+ \geq q(\gamma)$, so $W \subseteq \text{HCD}(q(\gamma))$ and λ is regular in W by Theorem 5.1; W is a ground by Input B. For the strong-limit clause R8: an injection $\lambda \rightarrow \mathcal{P}^W(\mu)$ in W with $\mu < \lambda$ would contradict $2^\mu < \lambda$ in V . The rest is Lemma 7.1 and Corollary 5.5. $\square$

Corollary 7.3 (Ground Axiom obstruction). [R1–R4, R6, R8] The theory $\text{ZFC} + \text{GA} + \exists \lambda \leq \gamma < \kappa [\text{CEx}_\gamma(\lambda) \wedge \text{SC}(\kappa)]$ is inconsistent. Under GA, every strongly compact cardinal is at most every cover bound of every cover-exacting cardinal; and $\text{ZFC} + \text{GA} + “\text{there is a proper class of strongly compact cardinals}”$ proves that no cardinal is cover exacting for any set-sized bound. No outer model of $\text{ZFC} + \text{GA}$ can preserve both $\text{CEx}_\gamma(\lambda)$ and a strongly compact $\kappa > \gamma$.

The inequality $\kappa > \gamma$, not merely $\kappa > \lambda$, is essential: for $\lambda < \kappa \leq \gamma$ monotonicity does not place $\text{HCD}(\kappa)$ inside $\text{HCD}(\gamma^+)$ R4. Nothing here excludes $\text{GA} + \text{CEx}_\gamma(\lambda)$ with a single strongly compact below λ : then $V = \text{HCD}(\delta)$ and the gap of Section 6 remains R3. The theorem is consonant with the known positive constructions, which are Prikry extensions.

Corollary 7.4 (General ground obstruction). [R1, R3, R7] Assume $\text{CEx}_\gamma(\lambda)$. If $W \models \text{ZFC}$ is an inner model with $W \subseteq \text{HCD}(q(\gamma))$ and $V = W[G]$, then the forcing is not λ -cc in W . If instead $V = W[G]$ for an arbitrary ground W with $|\mathbb{P}|^W \leq \nu < \lambda$, then W contains a cofinal $a \subseteq \lambda$ with $|a|^W \leq \nu$ and $a \notin \text{CD}(q(\gamma))$; so the assumption that all such small sets of W lie in $\text{CD}(q(\gamma))$ is contradictory.

Corollary 7.5 (Two strongly compacts). [R1, R6, R8] If $\delta < \lambda \leq \gamma < \kappa$ with $\text{SC}(\delta)$, $\text{SC}(\kappa)$, $\text{CEX}_\gamma(\lambda)$, then both $\text{HCD}(\kappa)$ and $\text{HCD}(\delta)$ are grounds and

$$\underbrace{\text{HCD}(\kappa)}_{\lambda \text{ inaccessible}} \subseteq \underbrace{\text{HCD}(q(\gamma))}_{\lambda \text{ regular}} \subsetneq \underbrace{\text{HCD}(\delta)}_{\text{cf}(\lambda) < \delta} \subseteq V,$$

the strict inclusion being witnessed on $[\lambda]^{<\delta}$. $\text{HCD}(\delta)$ has δ -approximation while $\text{HCD}(\kappa)$ fails it. Nothing prevents $\text{HCD}(\delta) = V$.

7.1 A successor stationary-set obstruction (second round)

Lemma 7.1 bounds a singularizing forcing at λ . With a strongly compact cardinal *below* λ the bound moves to the successor, and a specific stationary set must be lost. The mechanism (non-reflection after Prikry-type singularization versus reflection from strong compactness) is classical; R14 frees it from any assumption that V is a forcing extension.

Lemma 7.6 (Reflection). [R14; classical] If δ is strongly compact, $\theta > \delta$ is regular and $S \subseteq E_\omega^\theta$ is stationary, then $S \cap \alpha$ is stationary in some $\alpha < \theta$ with $\omega < \text{cf}(\alpha) < \delta$.

Proof. Let $j : V \rightarrow M$ come from a fine δ -complete ultrafilter on $\mathcal{P}_\delta(\theta)$ and $\beta = \sup j''\theta < j(\theta)$; then $\theta \leq \text{cf}^M(\beta) < j(\delta)$. Given a club $C \subseteq \beta$ in M , interleave C with $j''\theta$ and take $\alpha \in S$ closed under the interleaving function; $\text{cf}(\alpha) = \omega$ gives $j(\alpha) = \sup j''\alpha \in C \cap j(S)$. So M thinks $j(S)$ reflects at β ; apply elementarity. $\square$

Theorem 7.7 (Collapse or destroy). [R14] Let $W \subseteq V$ be an inner model of ZFC, λ a cardinal of V that is regular in W with $\text{cf}^V(\lambda) = \omega$, and let some $\delta < \lambda$ be strongly compact in V . Put $\theta = (\lambda^+)^W$ and $S = (E_\lambda^\theta)^W$. Then either $|\theta|^V = \lambda$, or $\theta = (\lambda^+)^V$ and the W -stationary set S is nonstationary in V .

Proof. Assume $\theta = (\lambda^+)^V$. S is stationary in W and $S \subseteq (E_\omega^\theta)^V$. If $\text{cf}^V(\alpha) > \omega$ then $\text{cf}^W(\alpha) < \lambda$, and a continuous cofinal sequence of length $\text{cf}^W(\alpha)$ in W is a club of α avoiding S ; so S reflects nowhere in V , and Lemma 7.6 makes it nonstationary. $\square$

Corollary 7.8 (Successor chain-condition bound). [R14] In the situation of Theorem 7.7, if $V = W[G]$ for $\mathbb{P} \in W$ then $\mathbb{P}$ is not $(\lambda^+)^W$ -cc in W ; every dense subset of $\mathbb{P}$ in W has W -size at least $(\lambda^+)^W$. (A θ -cc forcing preserves θ and every stationary subset of θ .)

Corollary 7.9. [R14] (a) If λ is exacting and some $\delta < \lambda$ is strongly compact, then $(\lambda^+)^{\text{HOD}}$ is collapsed or $(E_\lambda^{(\lambda^+)^{\text{HOD}}})^{\text{HOD}}$ is nonstationary in V . (b) If $\delta < \lambda \leq \gamma < \kappa$ with $\text{SC}(\delta)$, $\text{SC}(\kappa)$ and $\text{CEX}_\gamma(\lambda)$, then every presentation of V over $W = \text{HCD}(\kappa)$ has an antichain of W -size $(\lambda^+)^W$, and $(\lambda^+)^W$ is collapsed or $S = (E_\lambda^{(\lambda^+)^W})^W$ is nonstationary in V . If $(\lambda^+)^W = (\lambda^+)^V$, then S is already nonstationary in $\text{HCD}(\delta)$: some club $C \in \text{HCD}(\delta) \setminus W$ avoids it.

Proof. (a) $W = \text{HOD}$ and Input D(i). (b) Theorem 7.2 and Corollary 7.8. For the last sentence, V is an extension of $U = \text{HCD}(\delta)$ by a forcing of size below $(2^{2^\delta})^+ < \lambda$ (Input B; λ is a strong limit), which is θ -cc in U and so preserves stationary subsets of θ ; since S is nonstationary in V , it is nonstationary in U , and a club of U avoiding S cannot lie in W . $\square$

Part (b) strengthens the λ -cc bound of Theorem 7.2 to λ^+ when there is also a strongly compact cardinal below λ . It is an obstruction of the same conditional kind: it is incompatible with a (λ^+) -cc presentation, or with preservation of one named stationary set, not with the configuration itself.

8 Exacting cardinals: definable families of cofinal maps

This section uses only Input **D(i)**. Write $N = \text{HOD}_{V_\lambda}$. An OD_{V_λ} set need not belong to N , but an OD_{V_λ} set of ordinals, or graph between ordinals, does.

Theorem 8.1 (Projection width). [R9] Let λ be exacting, $0 < \mu < \lambda$, and let $F \in \text{OD}_{V_\lambda}$ be a nonempty set of cofinal maps $\mu \rightarrow \lambda$. Then for some $\xi < \mu$ the projection $P_\xi = \{f(\xi) : f \in F\}$ is unbounded in λ , and for every such ξ , $\text{ot}(P_\xi) = \lambda = |P_\xi|^V$; hence $|F|^V \geq \lambda$. If all $f \in F$ are increasing this holds from some coordinate onward.

Proof. All P_ξ are defined uniformly from one definition of F , so $\xi \mapsto \sup\{\beta + 1 : \beta \in P_\xi\}$ lies in N . If every P_ξ were bounded, regularity of λ in N would bound this function below λ , and then every $f \in F$ would be bounded. An unbounded $P_\xi \in N$ has order type λ , again by regularity. $\square$

No member of F is assumed definable and no choice is made inside N .

Corollary 8.2. [R9] If λ is exacting, there is no OD_{V_λ} family F with $0 < |F| < \lambda$ of cofinal maps from a fixed $\mu < \lambda$ into λ . If $c : \omega \rightarrow \lambda$ is cofinal, every $A \in \text{OD}_{V_\lambda}$ with $c \in A$ has $|A| \geq \lambda$; thus the principle “every $x \in V_{\lambda+1}$ lies in an OD_{V_λ} set of size below λ ” fails.

Proposition 8.3 (Fresh cofinal set). [R9; HOD form R2, R3, R8] If λ is exacting and $c \subseteq \lambda$ is cofinal of order type ω , then $c \notin \text{HOD}_{V_\lambda}$, but $c \cap a$ is finite and in HOD_{V_λ} for every $a \in \text{HOD}_{V_\lambda}$ that injects there into an ordinal below λ .

The cloud of Section 10.1 does not contradict Corollary 8.2: it may have size exactly λ , and its definition uses the generic sequence and a name, not parameters in V_λ .

8.1 Small families of short cofinal sets, bounded separation, and laws (second round)

Write C_λ for the set of cofinal $a \subseteq \lambda$ with $|a| < \lambda$ (equivalently $\text{ot}(a) < \lambda$); so $D_\lambda \subseteq C_\lambda$ in the notation of Section 9.

Lemma 8.4 (Fixed small sets). [R10, R12, R14–R18] Let $j : X \rightarrow V_\zeta$ be an exacting witness at λ with $\kappa = \text{crit}(j)$. (a) If $S \in X$, $S \subseteq \lambda$, $j(S) = S$ and $|S| < \lambda$, then $S \subseteq \kappa$. (b) There is no $\mathcal{F} \in X$ with $j(\mathcal{F}) = \mathcal{F}$, $\emptyset \neq \mathcal{F} \subseteq C_\lambda$ and $|\mathcal{F}| < \lambda$. The members of $\mathcal{F}$ may have different, unbounded, order types and need not be fixed.

Proof. (a) $\mu = |S|$ is a fixed ordinal below λ , so $\mu < \kappa$; a bijection $f : \mu \rightarrow S$ in X gives $j''S = \text{ran } j(f) = S$, and an order-preserving bijection of a set of ordinals onto itself is the identity. So S consists of fixed ordinals, which lie below κ by Lemma 3.1. (b) Let τ be the least order type of a member of $\mathcal{F}$ and $u = \bigcup\{b \in \mathcal{F} : \text{ot}(b) = \tau\}$. Then u is cofinal, $|u| \leq |\mathcal{F}| \cdot |\tau| < \lambda$, and u is definable from $\mathcal{F}$, hence fixed, contradicting (a). $\square$

The compression to the minimal order type R11 is what removes any uniform bound: a plain union could have size λ when the sizes of the members are unbounded. For subfamilies of D_λ the plain union R13, R16 or the coordinate projections R12, R14 serve equally.

Theorem 8.5 (No small definable family of short cofinal sets). [R10; parameter device R18] If λ is exacting, every nonempty $A \in \text{OD}_{V_\lambda}$ with $A \subseteq C_\lambda$ has $|A| \geq \lambda$.

Proof. Otherwise let ρ be the least rank of a parameter $p \in V_\lambda$ for which some counterexample lies in OD_p . At a sufficiently correct height, ρ is definable from λ , hence fixed by any witness j , hence $\rho < \text{crit}(j)$; so such a p lies in $V_{\text{crit}(j)}$ and is fixed. The least counterexample in the canonical well-order of OD_p is then fixed, contradicting Lemma 8.4(b). $\square$

The *minimal-rank-parameter device* of R18 is worth isolating: no ordinal or low-rank parameter of the original definition is assumed fixed, and no witness with a prescribed high critical point is needed. R10 and R14–R17 instead use Input D(ii) to raise the critical point above the rank of a given parameter; both routes are correct. Theorem 8.5 contains Input D(i) (take $A = \{a\}$), Corollary 8.2, and the version of the latter for maps with varying domains below λ R18.

Theorem 8.6 (Bounded-rank separation). [R10, R14, R17] Let λ be an uncountable cardinal with $|V_\alpha| < \lambda$ for all $\alpha < \lambda$. Then λ is regular in HOD_{V_λ} iff every $A \in \text{OD}_{V_\lambda}$ with $A \subseteq V_{\lambda+1}$ and $|A| < \lambda$ is separated at a bounded rank, i.e. $x \mapsto x \cap V_\alpha$ is injective on A for some $\alpha < \lambda$. In that case $A \in \text{HOD}_{V_\lambda}$, $A \subseteq \text{HOD}_{V_\lambda}$, and HOD_{V_λ} computes $|A|$ correctly. In particular this holds at every exacting λ , where for OD_{V_λ} sets $A \subseteq V_{\lambda+1}$: $|A| < \lambda$ iff A is separated at a bounded rank.

Proof. ($\Rightarrow$) The set of first-difference ranks $\{d(x, y) : x \neq y \in A\}$ is an OD_{V_λ} set of fewer than λ ordinals, hence bounded. Each $x \in A$ is then definable from A and $x \cap V_\alpha \in V_\lambda$. ($\Leftarrow$) If $b \in \text{HOD}_{V_\lambda}$ is short cofinal, $\{b \cap \beta : \beta \in b\}$ is not separated at any bounded rank. $\square$

The threshold is sharp ($A = \lambda$). The argument does not reach selectors on $[Q_\lambda]^n$, whose codes are subsets, not elements, of $V_{\lambda+1}$ R10.

Theorem 8.7 (Laws on cofinal sequences). [R10] Let $\text{cf}(\lambda) = \omega$ and let m be a countably additive probability measure on the σ -algebra on $S \subseteq D_\lambda$ generated by the sets $\{a \in S : a(n) < \alpha\}$. Then the medians $\min\{\alpha : m(\{a : a(n) < \alpha\}) > \frac{1}{2}\}$, $n < \omega$, form a cofinal set of order type ω , definable from S and m . Consequently, if λ is exacting there is no nonempty OD_{V_λ} family of fewer than λ such laws on D_λ .

Proof. The medians exist by continuity from below and are nondecreasing. If bounded by β , the sets $\{a : a(n) < \beta\}$ decrease to $\emptyset$ but all have measure above $\frac{1}{2}$, contradicting continuity from above. The family of median sets contradicts Theorem 8.5. $\square$

Countable additivity is essential: a nonprincipal ultrafilter on the tails of one sequence gives a finitely additive law with no medians R10.

9 Ultraexacting cardinals: failure of definable finite choice

For $\text{cf}(\lambda) = \omega$ let $D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega, \sup a = \lambda\}$, let $a =^* b$ mean $|a \Delta b| < \omega$, and $Q_\lambda = D_\lambda / =^*$. For $1 \leq r < n < \omega$ let $\text{Sel}_{n,r}(Q) = \{f : [Q]^n \rightarrow [Q]^r : f(B) \subseteq B\}$. The even/odd splitting of a critical sequence is in the Blue–Goldberg notes (Remark 2.4); the reports extend it in two complementary ways, through two different imported interfaces.

9.1 Finite cycles and finite families (interface: Input D(ii))

Lemma 9.1 (Finite algebra). [R5] Let τ be an N -cycle on A , σ a permutation of a finite set $\mathcal{F}$ with $|\mathcal{F}| = m$, and $v_f : [A]^n \rightarrow [A]^r$ with $v_f(B) \subseteq B$ and $v_{\sigma(f)}(\tau''B) = \tau''v_f(B)$. (a) If $N = n$, every σ -orbit has length divisible by $n/\gcd(n, r)$, and this bound is sharp. (b) If $N = n \cdot \text{lcm}(1, \dots, m)$, no such data exist.

Proof. (b) With $L = \text{lcm}(1, \dots, m)$, $\sigma^L = \text{id}$, so v_f commutes with τ^L , which acts on $B_0 = \{q_{kL} : k < n\}$ as a single n -cycle; a transitive cycle has no invariant subset of size $0 < r < n$. $\square$

Part (a) alone does *not* exclude finite families (two binary selectors can be swapped on a 2-cycle); the amplification (b) is the essential step.

Lemma 9.2 (Internal cycles). [R5] Let $j : X \rightarrow V_\zeta$ be an ultraexacting witness with $\zeta > \lambda + \omega$. Then the critical sequence $\vec{\kappa}$ of $e = j \upharpoonright V_\lambda$ belongs to X , and for every $N \geq 1$ the classes $q_i = [\{\kappa_{Nk+i} : k < \omega\}]$, $i < N$, lie in X and satisfy $j(q_i) = q_{i+1 \bmod N}$.

Proof. $\vec{\kappa}$ is definable from $e \in X$. Since enumerations with domain ω are in X , $j(a_i) = j''a_i$, which is a_{i+1} , or $a_0 \setminus \{\kappa_0\}$ for $i = N - 1$; the quotient removes exactly this defect. $\square$

Lemma 9.3 (Fixed canonical witnesses). [R5] If λ is ultraexacting, $p \in V_\lambda$, and some $z \in \text{OD}_p$ satisfies a first-order property $\Phi(z, p, \lambda, \vec{\kappa})$, then there are such a z_* and an ultraexacting witness j with $z_* \in X$ and $j(z_*) = z_*$.

Proof. Take the OD_p -least witness, reflect its definition, and use Input D(ii) to choose $\text{crit}(j) > \text{rank}(p)$. $\square$

Theorem 9.4 (No finite definable family of selectors). [R5] If λ is ultraexacting and $1 \leq r < n < \omega$, there is no nonempty finite $\mathcal{F} \in \text{OD}_{V_\lambda}$ with $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\lambda)$. More generally, no ultraexacting witness j has a nonempty finite $\mathcal{F} \in X$, $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\lambda)$, with $j(\mathcal{F}) = \mathcal{F}$.

Proof. For the second statement: $\mathcal{F} \subseteq X$ and $j \upharpoonright \mathcal{F}$ is a permutation σ ; restrict the selectors to the $n \cdot \text{lcm}(1, \dots, |\mathcal{F}|)$ -cycle of Lemma 9.2 and apply Lemma 9.1(b). The first statement follows by Lemma 9.3. Only permutations of finite sets are iterated; no iterability of j is used. $\square$

The family may be definable with no definable member. Note that $\text{Sel}_{n,r}(Q_\lambda)$ itself is ordinal definable and nonempty, yet has no nonempty finite OD_{V_λ} subset. R7 proves the singleton, ordinal-parameter case by a different route (Corollary 9.9).

Corollary 9.5. [R5, R7] At an ultraexacting λ there is no nonempty finite OD_{V_λ} family of linear orders of Q_λ , of tournaments on Q_λ , or of choice functions on its finite subsets; and no ordinal-definable injection of Q_λ into the ordinals.

Theorem 9.6 (Selector-coded Icarus enrichment). [R5] Let $N \models \text{ZF}$ be a transitive inner model with $V_{\lambda+1} \subseteq N$ and $j : N \rightarrow N$ elementary with $\text{crit}(j) < \lambda = \sup_n j^n(\text{crit}(j))$. Then no nonempty finite $\mathcal{F} \in N$, $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\lambda)$, has $j(\mathcal{F}) = \mathcal{F}$. For $f \in \text{Sel}_{n,r}(Q_\lambda)$ let $E_f \subseteq V_{\lambda+1}$ code f on representatives by flat codes $\{\langle 0, i \rangle\} \cup \{\langle k+1, \alpha \rangle : \alpha \in a_k\}$. Then there is no elementary $j : L(V_{\lambda+1}, E_f) \rightarrow L(V_{\lambda+1}, E_f)$ with $\text{crit}(j) < \lambda$ and $j(E_f) = E_f$.

This is a forbidden *information condition*, not a refutation of $\text{I0}^\#$ or of unspecified Icarus sets: nothing shows that a canonical enrichment defines such a selector. The quotient classes have rank $\lambda + 1$, which is why representatives must be coded.

9.2 Finite labels and transversals (interface: Input D(iii))

Let $D_m(\lambda)$ be the set of m -tuples from D_λ with pairwise distinct classes. A rule $L : D_m(\lambda) \rightarrow k$ is coded by the predicate $A_L = \{\langle i, \xi \rangle : \xi \in a_i\} \cup \{\langle m, t \rangle : t = L(\vec{a})\} \subseteq V_{\lambda+1}$, and every elementary e of $(V_{\lambda+1}, A_L)$ with $\text{crit}(e) < \lambda$ satisfies $L(e(\vec{a})) = L(\vec{a})$, using $e(a) = e''a$ for countable a R7.

Lemma 9.7 (Finite-permutation realization). [R7] For $e : V_{\lambda+1} \rightarrow V_{\lambda+1}$ with $\text{crit}(e) < \lambda$, every m , and every $\pi \in S_m$, there is a pairwise disjoint $\vec{a} \in D_m(\lambda)$ with $e(a_i) =^* a_{\pi(i)}$: on a cycle $(i_0 \dots i_{\ell-1})$ with offset t , put $a_{i_s} = \{\kappa_{\ell n+s} + t : n < \omega\}$.

Theorem 9.8 (Finite-label obstruction). [R7] Let λ be ultraexacting, F a finite S_m -set, and $L : D_m(\lambda) \rightarrow F$ ordinal definable, invariant under finite modification of each coordinate, and S_m -equivariant. Then every $\pi \in S_m$ fixes some element of F .

Proof. With e preserving A_L and $\vec{a}$ from Lemma 9.7: $L(\vec{a}) = L(e\vec{a}) = L(\pi^{-1} \cdot \vec{a}) = \pi^{-1} \cdot L(\vec{a})$. $\square$

Corollary 9.9. [R7] At an ultraexacting λ there is no ordinal-definable $s \in \text{Sel}_{m,r}(Q_\lambda)$ for $1 \leq r < m$ (an m -cycle fixes no proper nonempty subset of m).

The test does not ask for a label fixed by the whole group: S_3 acting on the disjoint union of its natural 3-point action and the 2-point sign action has a fixed point for each permutation but no common fixed point. R7 left open whether a rule L exists for such a target. R13 settles this: the elementwise test is also *sufficient* (Theorem 12.13).

Theorem 9.10 (No finite-valued transversal). [R7] If λ is ultraexacting there is no ordinal-definable $T \subseteq D_\lambda$ with $0 < |T \cap [a]| < \omega$ for every $a \in D_\lambda$.

Proof. Take e preserving T and $a = \{\kappa_n\}$, so $e(a) = a \setminus \{\kappa_0\}$. Then e injects the finite set $B = T \cap [a]$ into itself, so $e^p(b) = b$ for some $b \in B$, $p \geq 1$. Then $e^p \upharpoonright b$ is an increasing bijection of a set of order type ω , hence the identity, while e^p moves every ordinal in $[\text{crit } e, \lambda)$ and b is cofinal. $\square$

Remark 9.11 (Parameters in V_λ ; observation of the synthesis). Theorem 9.10 extends to $T \in \text{OD}_{V_\lambda}$ by the method of R5: Lemma 9.3 gives a fixed T_* and a witness j with $a = \text{ran}(\vec{\kappa}) \in X$ (Lemma 9.2); $B = T_* \cap [a]$ is a finite member of X with $j(B) = T_* \cap [j(a)] = B$, so j permutes $B \subseteq X$, and the same periodicity argument applies since $j(b) = j''b$ for $b \in B$.

Remark 9.12 (Controls). None of this contradicts Choice: every excluded object exists in V and merely fails to be definable. Modification invariance is essential, since D_λ itself carries an ordinal-definable lexicographic linear order R7. The argument does not apply to ordinary exacting cardinals, whose witnesses need not contain the critical sequence R5. And under $V_\lambda \subseteq \text{HOD}$ the quotient Q_λ^{bd} of bounded countable subsets of λ modulo finite difference lies in HOD and has an ordinal-definable well-order R5.

10 Consistency results

10.1 Cover exactingness from $\text{I3}_{\text{wf}}(0)$

Theorem 10.1. [R9] Let $j : V_\lambda \rightarrow V_\lambda$ be an $\text{I3}_{\text{wf}}(0)$ -embedding, $\kappa = \text{crit}(j)$, $U_0 = \{A \subseteq \kappa : \kappa \in j(A)\}$. Then $\mathbf{1}_{\mathbb{P}_{U_0}} \Vdash \text{CEX}_{\check{\kappa}}(\check{\kappa})$ for Prikry forcing $\mathbb{P}_{U_0}$. Consequently there is a countable transitive model of $\text{ZFC} + \exists \lambda \text{CEX}_\lambda(\lambda)$, and

$$\text{Con}(\text{ZFC} + \text{I3}_{\text{wf}}(0)) \implies \text{Con}(\text{ZFC} + \exists \lambda \text{CEX}_\lambda(\lambda)).$$

The proof combines three lemmas, the first two of which are reusable.

Lemma 10.2 (Enumerated-domain internalization). [R9] Let W be a transitive set model of enough of ZFC, $e : W \rightarrow W$ an *external* elementary map, λ a cardinal of W , and $c = \langle v_n \rangle \in W$ increasing cofinal in λ with $e(\lambda) = \lambda$, $e(v_n) = v_{n+1}$, $\omega < v_0$. Assume $V_\lambda \subseteq W$, $W \models |V_\lambda| = \lambda$, and $\alpha, Y \in W$ with $Y \subseteq V_\alpha^W$, $e(\alpha) = \alpha$, $e(Y) = Y$. Then W contains a relative λ -exacting witness for (V_α^W, Y) . It is not assumed that e or $e \upharpoonright V_\lambda$ is in W .

Proof. In W take $X < (V_\alpha^W, \in, Y)$ of size λ containing $V_\lambda \cup \{\lambda\}$, a bijection $f : \lambda \rightarrow X$, and put $Z = e(X)$, $g = e(f)$. The maps $e \upharpoonright f''v_n$ equal $g \circ (e \upharpoonright v_n) \circ f^{-1}$ and belong to W because $e \upharpoonright v_n \in V_\lambda$. In W form the tree of coherent finite lists of partial elementary maps $f''v_m \rightarrow g''v_{m+1}$ sending $\lambda \mapsto \lambda$, $v_0 \mapsto v_1$. It has an external branch, so by absoluteness of well-foundedness it has a branch in W , whose union is the required embedding $X \rightarrow Z < V_\alpha^W$. $\square$

Lemma 10.3 (Finite-change cloud). [R9] Let c be Prikry generic over a transitive N for a normal measure $D \in N$ on λ , $W = N[G_c]$, and $\tau \in N$ a name forced into V_α . Let $Y_{\tau,c} = \{\tau^{G_h} : h \text{ increasing, } |\text{ran } h \Delta \text{ran } c| < \omega\}$. Then $\tau^{G_c} \in Y_{\tau,c} \subseteq V_\alpha^W$, $Y_{\tau,c} \in W$, $|Y_{\tau,c}|^W \leq \lambda$, and any elementary $e : W \rightarrow W$ fixing $\lambda, \mathbb{P}_D^N, \tau, \alpha$ with $e(c) = c^-$ (the tail) satisfies $e(Y_{\tau,c}) = Y_{\tau,c}$.

The value τ^{G_c} itself may move; cover exactingness tolerates exactly this motion inside an invariant small family.

Lemma 10.4 (Localization). [R9, after ABGL Thm. 5.4] Let N_0 be transitive of size κ , $V_\kappa \cup \{\kappa, U_0 \cap N_0\} \subseteq N_0$, modelling enough of ZFC. With $J = j_{0,\omega} : N = J(N_0)$ is well-founded, there is an elementary $i : N \rightarrow N$ with $i \upharpoonright \Lambda = j \upharpoonright \Lambda$ and $i(J(a)) = J(a)$ for every $a \in N_0$, the critical sequence c is Prikry generic over N for $J(U_0 \cap N_0)$, and i lifts to $e : N[G_c] \rightarrow N[G_c]$ with $e(c) = c^-$.

Proof of Theorem 10.1. Otherwise, by cone homogeneity of Prikry forcing, some α and name τ have $1 \Vdash$ “no $Y \ni \tau$ of size $\leq \kappa$ admits a relative witness at α ”. Collapse a hull $H < V_\zeta$ of size κ containing $V_\kappa \cup \{U_0, \alpha, \tau\}$ to N_0 and apply Lemma 10.4; the images $a = J(\tilde{\alpha})$, $t = J(\tilde{\tau})$ are fixed by e . In $W = N[G_c]$ the cloud $Y_{t,c}$ is a set of size at most Λ containing t^{G_c} with $e(Y_{t,c}) = Y_{t,c}$ (Lemma 10.3), so Lemma 10.2 yields a relative witness in W , contradicting the transferred forcing statement. For the transitive model, note $V_\Lambda[G] = V[G]_\Lambda$ and pass to a countable elementary submodel of V_Λ . $\square$

Corollary 10.5 (Strength interval). [R9] In consistency strength, $I_2 > I_{3_{\text{wf}}}(0) \geq \exists \lambda \text{ CEx}_\lambda(\lambda) \geq \exists \lambda \text{ Ex}(\lambda) > I_3$; the strict comparisons are those of Input E. No equiconsistency is claimed, and the ultraexacting clause cannot be read into the construction.

This lowers the I_2 upper bound of Input C(iv). R9 notes that the Blue–Goldberg notes anticipate cover variants of related constructions, so no priority is claimed. The construction does not preserve a prescribed strongly compact below κ and so does not touch Problem 5.2.

Corollary 10.6 (Calibration of the barrier). [R1; with R6, R9] $\text{Con}(\text{ZFC} + I_2)$, and by Theorem 10.1 already $\text{Con}(\text{ZFC} + I_{3_{\text{wf}}}(0))$, implies

$$\text{Con}(\text{ZFC} + \exists \lambda [\text{CEx}_\lambda(\lambda) \wedge \text{cf}^V(\lambda) = \omega \wedge \text{cf}^{\text{HCD}(\lambda)}(\lambda) = \lambda]).$$

This adds no strength: the extra clauses are theorems about every such model (Theorem 5.1). Its role is to confirm that the barrier is a non-contradictory structural feature. The combination of R1, R6 and R9 in the displayed form is made here.

10.2 One I0-equiconsistent boundary theory

Five first-round reports, and all nine second-round reports, each derive an “equiconsistency package” from Input $\mathbf{D}(\text{iv}), (\text{v})$. Since every clause is a consequence of $\text{UEx}(\lambda) \wedge V_\lambda \subseteq \text{HOD}$, they merge into one statement.

Theorem 10.7 (Merged boundary package). [R2, R3, R5, R7, R8; R10–R18] Let T_∂ be ZFC plus the existence of a cardinal λ with $\text{UEx}(\lambda)$ and $V_\lambda \subseteq \text{HOD}$. Then $\text{Con}(T_\partial) \Leftrightarrow \text{Con}(\text{ZFC} + \text{I0})$, and T_∂ proves all of the following for that λ and every cofinal $c \subseteq \lambda$ of order type ω :

1. λ is the least exacting cardinal. [known: ABGL Cor. 3.10]
2. $V_\lambda^{\text{HOD}} = V_\lambda$; $\mathcal{P}(\alpha)^{\text{HOD}} = \mathcal{P}(\alpha)$ and $\text{cf}^{\text{HOD}}(\alpha) = \text{cf}(\alpha)$ for all $\alpha < \lambda$. [R3, R8]
3. $\text{cf}(\lambda) = \omega$ while $\text{HOD} \models “\lambda \text{ is strongly inaccessible}”$; $c \in V_{\lambda+1} \setminus \text{HOD}$, so λ is the first ordinal of cofinality or powerset disagreement and $\lambda + 1$ the first rank of disagreement. [R2, R3, R8]
4. $c \cap a$ is finite and in HOD whenever $a \in \text{HOD}$, $|a|^{\text{HOD}} < \lambda$; every $b \in \text{HOD}$ with $c \subseteq b$ has $|b|^V \geq \lambda$; so (HOD, V) fails θ -approximation and θ -covering for all uncountable $\theta \leq \lambda$. [R2, R3, R8]
5. $Q_\lambda^{\text{bd}} \in \text{HOD}$ has an ordinal-definable well-order, whereas the conclusions of Theorems 9.4, 9.8, 9.10 and Corollary 9.5 hold for Q_λ . [R5, R7]
6. Every conclusion of Section 12 holds, with OD in place of OD_{V_λ} since the two coincide: at least λ rigid classes q , for which moreover $H_q = \text{HOD}_{\{q\}} \models \text{ZFC}$ agrees with V below λ , regards λ as inaccessible, and contains no member of q ; full-width fibres for every OD complete section; the exact finite-label classification, including an OD choice-or-orientation rule on triples although neither pure rule is OD; no OD probability kernel. The bounded quotient has an OD transversal. [R10–R18]

Proof. Equiconsistency is Input $\mathbf{D}(\text{iv}), (\text{v})$. For (1): a smaller exacting μ would have a cofinal ω -sequence of rank $\mu < \lambda$, hence in HOD , contradicting Input $\mathbf{D}(\text{i})$. For (2): the relevant subsets and cofinal maps have rank below λ . (3),(4): HOD -regularity is Input $\mathbf{D}(\text{i})$; strong limit as in Theorem 7.2; the rest as in Corollary 5.5. (5): Remark 9.12 and Section 9. (6): Section 12; under $V_\lambda \subseteq \text{HOD}$ every parameter in V_λ , in particular a well-order of the reals, is ordinal definable. $\square$

The equiconsistency compares consistency statements; it does not say that I0 holds at the same ordinal, and it is not a new proof of the ABGL comparison. Its use is as a *control*: total agreement with HOD below λ does not supply the one countable sequence that would be fatal, so no negative argument may import covering at λ from rank agreement below λ .

11 Class embeddings: the internal unbounded-range obstruction

Let $j : V \rightarrow M$ be a nontrivial elementary embedding into a transitive inner model (in a class theory where the needed restrictions exist), $\kappa = \text{crit}(j)$, and let $\theta > \kappa$ be regular uncountable with $j(\theta) = \theta$. For $\lambda = \sup_n j^n(\kappa)$, $\theta = \lambda^+$ is such a cardinal *without* any cardinal-correctness assumption: $j(\theta) = (\lambda^+)^M \leq \lambda^+ \leq j(\theta)$ R3. Let $C_j = \{\beta < \theta : j''\beta \subseteq \beta\}$, a club whose points of cofinality ω are fixed by j ; fix a partition $\vec{S}$ of E_ω^θ into θ stationary sets and let $\vec{T} = j(\vec{S})$.

Lemma 11.1 (Stationarity spectrum; known). [HKP; proof in R3] $\{\xi < \theta : T_\xi \text{ is stationary in } V\} = j''\theta$, and the single club C_j is disjoint from every T_ξ with $\xi \notin j''\theta$.

R2, R6 and R7 each examined this reconstruction as a candidate result and rejected it as already contained in Hamkins–Kirmayer–Perlmutter [8, Thm. 11–12]; R3 attributes it the same way and adds:

Theorem 11.2 (No internal unbounded part of the range). [R3] No $A \in M$ with $A \subseteq j''\theta$ is unbounded in θ .

Proof. Let $f \in M$ enumerate such an A increasingly. Since $A = j''B$ with B unbounded, $f(\xi) = j(b_\xi) \geq j(\xi)$. So the M -club $D_f = \{\beta : f''\beta \subseteq \beta\}$ is contained in C_j . But T_κ is stationary in M and $\kappa \notin j''\theta$, so $D_f \cap T_\kappa \neq \emptyset = C_j \cap T_\kappa$. $\square$

Corollary 11.3. [R3] Every unbounded $B \subseteq \theta$ in M contains θ many labels ξ with T_ξ stationary in M but not in V . The ambient nonstationary ideal on θ is not amenable to M . (The set of surviving labels is nevertheless stationary in V : abundance in V and availability in M are different.)

Remark 11.4 (A sanity control added in the synthesis). The proof shows that M contains no club subset of C_j , so no function in M dominates $j \restriction \theta$. One may check this on the ultrapower by a normal measure on κ with $\theta > \kappa$ inaccessible. The M -definable set of M -strong-limit cardinals of cofinality ω is *not* contained in C_j : if $\alpha > \kappa$ is a V -strong limit of cofinality κ , then $\sup j''\alpha = \alpha < j(\alpha)$ and $j(\alpha)$ is an M -strong limit, so the least M -strong limit β' above α satisfies $\alpha < \beta' \leq j(\alpha)$ and is not closed under j . The familiar fixed points (the V -strong limits of cofinality ω) do not form a set in M .

The result is a conditional inconsistency for a strengthened embedding theory. It uses no cardinal correctness, so the global cardinal-preserving problem of the research plan still lacks the implication from $\text{Card}^M = \text{Card}^V$ to any internal selection of externally stationary cells.

12 Sections of the cofinal quotient

This section merges the second round of reports. Throughout, λ has cofinality ω , D_λ is the set of cofinal subsets of λ of order type ω , $a =^* b$ means $|a \Delta b| < \omega$, $[a]$ is the class of a and $Q_\lambda = D_\lambda / =^*$, as in Section 9; $C_\lambda \supseteq D_\lambda$ is the set of short cofinal subsets of λ (Section 8.1). A *complete section* is a set $T \subseteq D_\lambda$ meeting every class; it is *thin* if $|T \cap q| < \lambda$ for every q , and a *transversal* if $|T \cap q| = 1$. No bound below λ uniform in q is ever assumed. $\text{OD}_{V_\lambda \cup \{q\}}$ allows ordinals, finitely many parameters in V_λ , and the class q as *one* set parameter — not its members. All cardinalities are computed in V .

Every class has exactly λ members ($d \mapsto a \Delta d$ is a bijection from $[\lambda]^{<\omega}$), so λ is the largest possible fibre; and $|Q_\lambda| = 2^\lambda$ for a strong limit λ of cofinality ω R15.

12.1 Fixed classes

Lemma 12.1 (Orbits and offsets). [R12, R15; R13, R14, R17; R10, R16, R18] Let $j : X \rightarrow V_\zeta$ be an ultraexacting witness with $\zeta > \lambda + \omega$, $e = j \restriction V_\lambda \in X$, $\kappa = \text{crit}(j)$, $\kappa_n = e^n(\kappa)$.

1. The set of *roots* $R_e = [\kappa, \lambda) \setminus e''[\kappa, \lambda)$ has size λ , and the forward orbits $a_r = \{e^n(r) : n < \omega\}$, $r \in R_e$, partition $[\kappa, \lambda)$ into members of D_λ .
2. Each a_r and $q_r = [a_r]$ belong to X , and $j(a_r) = a_r \setminus \{r\}$, so $j(q_r) = q_r$. The classes q_r are pairwise distinct.
3. In particular this holds for the critical sequence ($r = \kappa$) and for the offsets $a_t = \{\kappa_n + t : n < \omega\}$, $t < \kappa$.

Proof. e maps $[\kappa_{n-1}, \kappa_n)$ into $[\kappa_n, \kappa_{n+1})$, which has size κ_{n+1} , so κ_{n+1} roots lie there. Every ordinal in $[\kappa, \lambda)$ has only finitely many e -predecessors (Lemma 3.1), hence lies on a unique orbit. Orbits are definable from $e \in X$ and $r \in V_\lambda \subseteq X$; enumerations with domain ω are mapped pointwise. $\square$

This is the only place where ultraexactingness, rather than exactingness, is used in this section: an exacting witness need not contain any fixed class. All nine reports record the question whether exactingness suffices as open.

12.2 Rigid classes

Definition 12.2 (Rigidity). [R18] A class $q \in Q_\lambda$ is *rigid*, $\text{Rig}(\lambda, q)$, if every nonempty $A \in \text{OD}_{V_\lambda \cup \{q\}}$ with $A \subseteq C_\lambda$ has $|A| \geq \lambda$.

Theorem 12.3 (Rigid classes exist). [R18; with R12, R15, R17] Let λ be ultraexacting and let $j : X \rightarrow V_\zeta$ be an ultraexacting witness at a sufficiently correct height. Then every $q \in X \cap Q_\lambda$ with $j(q) = q$ is rigid. Consequently at least λ classes, with pairwise disjoint representatives, are rigid; and the bound in Definition 12.2 is attained, by $A = q$.

Proof. Exactly as for Theorem 8.5, with q as an additional fixed parameter: minimize the rank of the V_λ -parameter (so that it falls below $\text{crit}(j)$ and is fixed), take the least counterexample in the canonical well-order, and contradict Lemma 8.4(b). Lemma 12.1 supplies λ fixed classes. $\square$

R18 proves rigidity of the critical-sequence class; R17 obtains λ classes that are saturated for all OD_{V_λ} sets simultaneously, using instead a bijection $b : \lambda \rightarrow V_\lambda$ in X with $j(b) = b$ (so that $\text{OD}_{V_\lambda} \subseteq \text{OD}_b$) and a uniform bounded-rank correctness lemma. Applying the device of R18 to each orbit class of R12, R15 gives both conclusions at once; this combination is made here.

Corollary 12.4 (What rigidity gives). [R11, R15, R17, R18] Let q be rigid.

1. *Zero-or-full traces.* $|T \cap q| \in \{0, \lambda\}$ for every $T \subseteq D_\lambda$ in $\text{OD}_{V_\lambda \cup \{q\}}$.
2. *Small families.* The same holds for every member T of any family $\mathcal{T} \subseteq \mathcal{P}(D_\lambda)$ in $\text{OD}_{V_\lambda \cup \{q\}}$ with $|\mathcal{T}| < \lambda$, and for all values $F(x, \alpha)$ of a definable $F : V_\lambda \times \theta \rightarrow \mathcal{P}(D_\lambda)$. The members need not be definable.
3. *Cofinal summaries.* If $R \in \text{OD}_{V_\lambda \cup \{q\}}$ assigns to each class a subset of C_λ , then $R(q) = \emptyset$ or $|R(q)| \geq \lambda$.
4. *The model $N_q = \text{HOD}_{V_\lambda \cup \{q\}}$.* $\text{cf}^{N_q}(\lambda) = \lambda$; $q \notin N_q$ and $q \cap N_q = \emptyset$; every $\text{OD}_{V_\lambda \cup \{q\}}$ set containing a member of q has size at least λ ; and the projection-width theorem 8.1 holds relative to q .

Proof. (1) $T \cap q$ is a definable subfamily of $D_\lambda \subseteq C_\lambda$. (2) For $T \in \mathcal{T}$ with $0 < |T \cap q| < \lambda$ the union $u_T = \bigcup (T \cap q)$ is cofinal of size below λ ; the set of all such u_T is a definable subfamily of C_λ of size at most $|\mathcal{T}| < \lambda$, hence empty. Each value $F(x, \alpha)$ is itself definable from x and α , so (1) applies to it. (3), (4) are instances of the definition. $\square$

Corollary 12.5 (Sharp width of definable sections). [R10–R18] Let λ be ultraexacting.

1. Every complete section $T \in \text{OD}_{V_\lambda}$ has at least λ classes with $|T \cap q| = \lambda$, and one set of λ classes serves all such T R17. For arbitrary $T \subseteq D_\lambda$ in OD_{V_λ} , at least λ fibres are empty or full R13.
2. There is no thin complete section in OD_{V_λ} : the *definable width* $\min\{\sup_q |T \cap q| : T \in \text{OD}_{V_\lambda} \text{ complete}\}$ is exactly λ . This contains Theorem 9.10.

3. No nonempty OD_{V_λ} family of fewer than λ thin complete sections exists (in particular no countable definable family of transversals), and no such family of thin *partial* sections has complete union R15. Every OD_{V_λ} set having a thin complete section as a member has size at least λ R12, R15.
4. Even the local demand fails: it is not the case that every class q has some $T \in \text{OD}_{V_\lambda}$ with $0 < |T \cap q| < \lambda$ R17.

Remark 12.6 (Sharpness). (a) $T = D_\lambda$ is ordinal definable with all fibres of size λ ; thin transversals exist by Choice. (b) The parameter rank cannot be raised: no class is saturated for all $\text{OD}_{V_{\lambda+1}}$ sets (use $T = \{a\}$, $a \in q$) R17. (c) The family bound cannot be raised in the local lemma: a fixed class is itself a fixed family of size λ R10. (d) Not every class is rigid: for each increasing cofinal c there is a redundantly coded class q_c from which c is ordinal definable, so $\text{cf}^{N_{q_c}}(\lambda) = \omega$ R11. (e) For every class q there are λ distinct classes q_i mutually ordinal definable with q , hence with $N_{q_i} = N_q$; so λ distinct rigid classes can share one model R11.

12.3 The structure of a full fibre

For $a =^* b$ let $\text{ind}_a(b) = |b \setminus a| - |a \setminus b| \in \mathbb{Z}$; then $\text{ind}_a(c) = \text{ind}_a(b) + \text{ind}_b(c)$ and $\text{ind}_{a \setminus \{\min a\}} = \text{ind}_a + 1$. For $B \subseteq [a]$ put $\text{Spec}_a(B) = \{\text{ind}_a(b) : b \in B\}$ and $P_n(B) = \{b(n) : b \in B\}$.

Theorem 12.7 (Coordinate width and full phase). [R12] Let j be an ultraexacting witness, $a \in D_\lambda \cap X$ with $j(a) = a \setminus \{\min a\}$, and $\emptyset \neq B \subseteq [a]$ with $B \in X$, $j(B) = B$. Then there is n_0 such that for every $r \in \mathbb{Z}$ and $n \geq n_0$ the phase $B_r = \{b \in B : \text{ind}_a(b) = r\}$ has $P_n(B_r)$ cofinal in λ of size λ . In particular $\text{Spec}_a(B) = \mathbb{Z}$ and every phase has size λ . Hence every complete section $T \in \text{OD}_{V_\lambda}$ has these properties on at least λ classes.

Proof. $\text{Spec}_a(B) = \text{Spec}_{j(a)}(j(B)) = \text{Spec}_a(B) + 1$, so the nonempty spectrum is $\mathbb{Z}$. The cardinals $\theta_r = |B_r|$ satisfy $j(\theta_r) = \theta_{r-1}$; a two-sided orbit of ordinals $\leq \lambda$ is constant, with value λ or below κ , and the latter would make B a fixed family of size below λ . A fixed small $P_n(B)$ would lie below κ by Lemma 8.4(a). $\square$

Corollary 12.8. [R12] At an ultraexacting λ no complete section in OD_{V_λ} (i) misses an integer phase in every class, or a residue class of phases modulo m ; (ii) has in every class a pointwise bound $g_q : \omega \rightarrow \lambda$ on its members; or (iii) has in every class a finite bound on the number of points inserted, or deleted, relative to some reference representative. The two conditions are independent: there is a complete section with all fibres of size λ and a single phase, and one with countable fibres and full spectrum.

Corollary 12.9 (Colourings, coarser relations, homomorphisms). [R14–R18] Let λ be ultraexacting. (a) For every rigid q , every $f : D_\lambda \rightarrow \text{Ord}$ in OD_{V_λ} and every ξ , $|q \cap f^{-1}\{\xi\}| \in \{0, \lambda\}$; so there is no OD_{V_λ} assignment of a well-order to each class that is constant on classes. (b) If $E \in \text{OD}_{V_\lambda}$ is an equivalence relation on D_λ with $a E (a \setminus \{\min a\})$ for all a , every OD_{V_λ} set meeting every E -class meets some E -class in at least λ points. (c) If $E \in \text{OD}_{V_\lambda}$ has all classes of size below λ , there is no nonempty OD_{V_λ} family of fewer than λ maps $f : D_\lambda \rightarrow D_\lambda$ with $a =^* b \Rightarrow f(a) E f(b)$; for instance when E is the orbit relation of a definable action of a group of size below λ .

12.4 Rules with cofinal output and randomized selection

Theorem 12.10. [R10, R11, R16, R18] If λ is ultraexacting, there is no nonempty OD_{V_λ} family of fewer than λ rules R on Q_λ with $\emptyset \neq R(q) \subseteq C_\lambda$ and $|R(q)| < \lambda$ for every q . The outputs need not represent, or be related to, the input class. Equivalently there is no such tail-invariant rule on D_λ ; in particular no OD_{V_λ} map $Q_\lambda \rightarrow C_\lambda$.

Proof. Let q be rigid and let $u(B)$ denote the least-order-type compression of Lemma 8.4. For a family $\mathcal{F}$ as described, $\{u(R(q)) : R \in \mathcal{F}\}$ is a nonempty subfamily of C_λ of size below λ , definable from $\mathcal{F}$ and q , contradicting rigidity. $\square$

Corollary 12.11 (No definable randomized selection). [R10] If λ is ultraexacting there is no nonempty OD_{V_λ} family of fewer than λ *probability kernels*, i.e. assignments to each class q of a countably additive probability law on q (on the σ -algebra of Theorem 8.7). Dirac kernels exist by Choice.

Proof. Theorem 8.7 converts a kernel into the rule $q \mapsto \text{median set of } K(q)$. $\square$

12.5 The exact finite-label classification

In the setting of Theorem 9.8, let $\Gamma \leq S_m$ act on a finite nonempty set F , call a rule $L : D_m(\lambda) \rightarrow F$ *admissible* if it is invariant under finite modification of the coordinates and Γ -equivariant, and let $\text{CF}(\Gamma, F)$ say that every single $g \in \Gamma$ fixes some element of F .

Lemma 12.12 (Cyclic stabilizers). [R13] To $\vec{a} \in D_m(\lambda)$ assign its *incidence word* $w_{\vec{a}}(n) = \{i < m : \delta_n \in a_i\}$, where $\langle \delta_n \rangle$ enumerates $\bigcup_i a_i$; finite modifications change w only up to *shift-tail equivalence* ($w(r+n) = v(s+n)$ for all n), and $w_{g \cdot \vec{a}} = g \cdot w_{\vec{a}}$. For every word w in which each label occurs infinitely often and any two labels are separated infinitely often, the stabilizer in Γ of the shift-tail class of w is cyclic.

Proof. $G_w = \{(d, g) \in \mathbb{Z} \times \Gamma : w(n+d) = g \cdot w(n) \text{ for all large } n\}$ is a subgroup whose projection to $\mathbb{Z}$ is injective (if $(0, g) \in G_w$ and g moves i , the labels i and $g^{-1}i$ are not separated infinitely often). So G_w is cyclic, and the stabilizer is its image in Γ . $\square$

Theorem 12.13 (Classification). [R13; necessity R7] Let λ be ultraexacting. An admissible rule $L : D_m(\lambda) \rightarrow F$ in OD_{V_λ} exists if and only if $\text{CF}(\Gamma, F)$. For sufficiency ultraexactingness is not needed: for every uncountable λ of cofinality ω the rule is definable from a well-order of $\mathcal{P}(\omega)$, a parameter in $V_{\omega+10}$; it is OD if the reals have an OD well-order, in particular if $V_\lambda \subseteq \text{HOD}$.

Proof. Necessity is Theorem 9.8. For sufficiency, the rule factors through shift-tail classes of words, which are coded by reals. In each Γ -orbit of classes take the class of the least coded word v ; its stabilizer is cyclic (Lemma 12.12), a generator fixes some label by CF, and so the whole stabilizer fixes the least such label f ; put $L(g \cdot v) = g \cdot f$. $\square$

Corollary 12.14. [R13] (a) At an ultraexacting λ there is an OD_{V_λ} rule assigning to every unordered triple of classes *either* a distinguished member *or* a cyclic orientation, although no OD_{V_λ} rule always does the first and none always does the second; every such mixed rule uses each output type on at least λ triples. (b) The five-element S_3 -set of Section 9 is, up to isomorphism, the smallest S_3 -set satisfying CF without a global fixed point. (c) A finite group admits a finite set with CF and no global fixed point iff it is not cyclic. (d) If strictly increasing maps f satisfy $f''a_i =^* a_{\pi_f(i)}$ on one common tuple, the π_f generate a cyclic group: several embeddings cannot realize a noncyclic relabelling group on one tuple, which is why the elementwise test cannot be upgraded to a whole-group test.

This settles the question left open in the first edition after Theorem 9.8, and shows that the obstruction of Section 9 is exactly the cyclic one.

12.6 Preserved-predicate versions

Theorem 12.15. [R10–R13, R15–R18] (a) Let $e : (V_{\lambda+1}, \in, T) \rightarrow (V_{\lambda+1}, \in, T)$ be elementary with $\text{crit}(e) < \lambda$ and $T \subseteq D_\lambda$. Then $|T \cap [a]| \in \{0, \lambda\}$ for the critical-sequence set a and for each offset a_t , $t < \text{crit}(e)$; so T is not a thin complete section. (b) Let $N \models \text{ZF}$ be a transitive inner model with $V_{\lambda+1} \subseteq N$ and $j : N \rightarrow N$ elementary with $\text{crit}(j) < \lambda = \sup_n j^n(\text{crit}(j))$ and set-sized restrictions. If $\mathcal{T} \in N$, $j(\mathcal{T}) = \mathcal{T}$, $\mathcal{T} \subseteq \mathcal{P}(D_\lambda)$ and $|\mathcal{T}|^V < \lambda$, then $|T \cap q|^V \in \{0, \lambda\}$ for all $T \in \mathcal{T}$ and all λ orbit classes q ; likewise Theorem 12.7 holds for a fixed complete section, and no $f : Q_\lambda \rightarrow D_\lambda$ in N is fixed. No Choice in N is used, and all sizes are ambient. (c) Hence for a thin (or phase-proper) complete section T , or for the code A_f of a map $f : Q_\lambda \rightarrow D_\lambda$, there is no elementary $j : L(V_{\lambda+1}, T) \rightarrow L(V_{\lambda+1}, T)$ with $\text{crit}(j) < \lambda$ and $j(T) = T$.

Proof. (a) The fibre is a subset, not an element, of $V_{\lambda+1}$; one applies e to the elements $\bigcup(T \cap [a])$ R13 or $P_n(T \cap [a])$ R16 of $V_{\lambda+1}$, which are fixed and would be small. (b), (c) as in Theorems 9.6 and 12.3, the orbits now being available because $V_{\lambda+1} \subseteq N$. $\square$

These extend Theorem 9.6 from finite families of selectors to thin sections. As there, they are forbidden *information conditions*: no report shows that a canonical Icarus enrichment defines a thin section.

12.7 Periodic classes of one embedding

Proposition 12.16. [R15] For $e : V_\lambda \rightarrow V_\lambda$ let $\bar{e}([a]) = [e''a]$. A class is fixed by $\bar{e}^r$ iff it has a representative that is a finite nonempty union of forward e^r -orbits in $[\kappa, \lambda)$. There are exactly λ classes of each least period $r \geq 1$, hence exactly λ periodic classes, out of 2^λ ; every periodic class lies in X , and Corollary 12.4(1),(2) holds at each of them for families fixed by j .

So a single witness cannot give more than λ test classes; whether a definable complete section must have more than λ full fibres is open.

13 Comparison of the reports

13.1 Agreement

The seven group-A reports agree on every mathematical conclusion they share. All nine agree on status: no unconditional inconsistency, no resolution of Problem 5.2, no priority claim, no formal verification. No contradiction between any two reports was found.

13.2 Differences that matter

Topic	Finding
Dependence on lecture notes	R1–R3, R8 import the stationary characterization (Prop. 3.4) and prove a focused transfer lemma; R7 imports both Lemma 3.3 and Prop. 3.4; R4 and R6 prove persistent covers, exact hulls and general transfer from the definition. Route H therefore removes the barrier's dependence on unrefereed results other than definitions, and should be preferred.
Threshold	$\eta > \gamma$ (R1, R3, R8), γ^+ (R2, R4, R7), $q(\gamma)$ (R6). Only R6 notices the singular endpoint, which gives the clean statement " $\text{CEx}_\lambda(\lambda) \Rightarrow \lambda$ regular in $\text{HCD}(\lambda)$ ".

Topic	Finding
Extra hypothesis below λ	Seven formulations P_1 – P_7 ; all are instances of one argument, and the two size conventions coincide (Corollary 6.2). The weakest uniform principles are P_4 (R2) and P_5 (R6), which are not comparable on their face.
Forcing lower bound	R1, R2 bound $ \mathbb{P} $; R3, R4, R6, R8 prove failure of the λ -chain condition, which is stronger and also yields density $\geq \lambda$. R4 stresses the bound is λ , not λ^+ .
Ground profile	Only R8 proves that λ is strongly inaccessible (not merely regular) in the ground; the same argument is reused for HOD in Theorem 10.7(3).
Finite choice	R5 and R7 use different interfaces. R5 (witnesses with high critical point and $j \restriction V_\lambda \in X$) gets finite families and parameters in V_λ ; R7 (one preserved OD predicate on $V_{\lambda+1}$) gets the general label theorem and the transversal theorem, with ordinal parameters only. R7's selector corollary is the singleton case of R5's theorem; R7's other results are not covered by R5.
I0 packages	Five overlapping packages (Bdry, T_{first} , Profile, T_1 , T_{bd}); all are corollaries of $\text{UEx}(\lambda) \wedge V_\lambda \subseteq \text{HOD}$ and merge into Theorem 10.7.
Stationary track	Consistent: three reports discard the HKP reconstruction as known; R3 also credits it to HKP and contributes only the strengthening, Theorem 11.2.
Citations	The UEx/I0 equiconsistency is cited as “Theorem A” (R3, R7) and “Theorem 4.5” (R2, R5, R8) of [5]; the δ -cover theorem as Thm. 4.14, with Cor. 4.7 added in R7; the trace argument as “proof of Thm. 3.7” or “p. 10” of [2]. These appear to be the same items under introduction and body numbering, but were not re-checked against the sources here.

13.3 The second round

The nine reports R10–R18 were written independently from the first edition of this synthesis, and all nine prove Corollary 12.5(2). No contradiction among them, or with the first round, was found. They differ as follows.

Topic	Finding
Local lemma	Three equivalent devices turn a small fixed family into a small fixed set of ordinals: the union (R13, R16; only for families of countable sets), coordinate projections (R12, R14), and compression to the least order type (R10, R11, R18; works for arbitrary short cofinal sets). Only the last gives Lemma 8.4(b) in full.
Fixing the definable object	R10, R12–R17 raise the critical point above the rank of the parameter (Input D(ii)); R11, R13 use the predicate form (Input D(iii)); R18 minimizes the rank of the parameter, which needs neither and is uniform in the parameter. R17 achieves uniformity differently, through a fixed bijection $\lambda \rightarrow V_\lambda$.
How many classes	One (R10, R11, R16, R18); κ offsets per witness, hence λ (R13, R14, R17); λ orbits of one witness (R12, R15). R15 shows λ is the most one witness can give.
Strength of the statement	Single section (R11, R13, R16); small families of complete sections (R10, R12 with a uniform bound, R14, R17); small families of partial sections (R15); all OD_{V_λ} sets at once (R17); definability from q itself (R11 for single sets, R18 for families). Theorem 12.3 and Corollary 12.4 are the common upper bound. R12's Proposition on small clouds is the weakest form and is subsumed.
Beyond cardinality	Only R12 analyses the inside of a full fibre (coordinates, phases). Only R13 treats finite labels, and it is the only second-round report that answers a question posed in the first edition. Only R14 returns to cover exactingness. Only R10 treats measures.
I0 packages	Nine more, all corollaries of $\text{UEx}(\lambda) \wedge V_\lambda \subseteq \text{HOD}$; merged into Theorem 10.7(6).

Topic	Finding
Relation to the Lean development	The abstract theorem <code>no_finite_valued_transversal</code> is the finite case of Lemma 8.4(b); the second-round results are not yet formalized.

13.4 Relative reliability

The group-A, B and E arguments are short and were re-derived during the merge without difficulty; the only external mathematics they use are Inputs A, B and D(i). The group-C arguments are elementary once the two imported interfaces are granted; R5 and R7 include Python scripts that exhaustively check only the finite algebra (Lemma 9.1, Lemma 9.7, the S_3 example) and are not evidence for the set-theoretic steps. Theorem 10.1 is the most delicate result: it is the only new forcing construction, it depends on the precise form of the ABGL well-founded-part localization (Input E, Lemma 10.4), and it appears in a single report. Its two new lemmas (10.2, 10.3) were re-derived here and appear sound; the imported interface was not checked. It should be the first target of specialist review. The second-round arguments are short and uniform: everything rests on Lemma 8.4, Lemma 12.1 and the canonical-counterexample device, all re-derived here. The two places that needed care are the choice of a height correct for the definability class *before* the witness (and hence the fixed class) is chosen, which R17 and R18 treat explicitly, and the rank bookkeeping in Theorem 12.15(a). Lemma 12.12 is elementary and was checked in full. R12 and R13 include Python checks of finite combinatorics only.

14 What remains open

Question 14.1 (Promotion; all group-A reports). Assume $\delta < \lambda \leq \gamma$, $SC(\delta)$, $CE_{\gamma}(\lambda)$. Do these hypotheses (or a named strengthening of strong compactness short of extendibility) imply P_4 , P_5 , or P_7 for one of the sets a of Theorem 6.1? A positive answer refutes the configuration of Problem 5.2. The entire unproved step is raising the completeness of an ultrafilter code from δ to $q(\gamma)$; strong compactness extends filters at their given completeness and does not do this.

Question 14.2 (Positive side). Can a cover-exacting construction preserve a prescribed strongly compact δ below the cover-exacting cardinal? Any such model must exhibit the separation of Theorem 6.1, must fail P_1 – P_6 , and, if it has a strongly compact above γ , must have the proper ground of Theorem 7.2; it cannot satisfy GA in that case.

Question 14.3 (R9). Can the $I3_{wf}(0)$ upper bound for $\exists \lambda \ CE_{\lambda}(\lambda)$ be lowered, or does cover exactingness carry an iteration lower bound beyond that of exactingness? Under a smaller strongly compact, is there an $OD_{V_{\lambda}}$ family of fewer than λ candidates containing a short cofinal map (which would contradict Corollary 8.2)?

Question 14.4 (R5, R7). Does ordinary exactingness already exclude an $OD_{V_{\lambda}}$ binary selector on Q_{λ} ? What is the least size of a nonempty $OD_{V_{\lambda}}$ subfamily of $Sel_{n,r}(Q_{\lambda})$ at an ultraexacting λ ; can it be countable? Does a canonical Icarus enrichment define a preserved selector, thin section, or map $Q_{\lambda} \rightarrow D_{\lambda}$ (Theorem 12.15)? (The first edition also asked whether a modification-invariant equivariant rule can exist for a finite S_m -set with elementwise but no global fixed points; Theorem 12.13 answers yes.)

Question 14.5 (R10–R18). (i) Does ordinary exactingness exclude an $OD_{V_{\lambda}}$ thin complete section? The local lemma needs only exactingness; the missing step is a fixed class in the domain. (ii) Can the thin complete sections, the kernels, or the maps $Q_{\lambda} \rightarrow D_{\lambda}$ have a nonempty $OD_{V_{\lambda}}$ subfamily of size

exactly λ ? (iii) Must an OD_{V_λ} complete section have more than λ full fibres, perhaps 2^λ ? (iv) Is there an intrinsic combinatorial characterization of the rigid classes, and are there more than λ of them, or more than one model N_q ? (v) In Corollary 7.9(b), must $(\lambda^+)^{\text{HCD}(\kappa)}$ be collapsed?

Question 14.6 (R3). Does $\text{Card}^M = \text{Card}^V$ for $j : V \rightarrow M$ yield an unbounded subset of $j''\theta$ in M , or amenability of the ambient nonstationary ideal? Either would refute cardinal-preserving embeddings by Theorem 11.2.

Not addressed by any report: ordinary exacting cardinals above an extendible, the HOD Conjecture fork, HOD-Berkeley cardinals, and the choiceless hypotheses of the research plan.

A Concordance

• = stated and proved in the report; ◦ = used, imported, or proved only in a weaker or special form.

Result	R1	R2	R3	R4	R5	R6	R7	R8	R9
Lemma 3.1 critical-sequence normalization	•	•	•	•	◦	•	•	•	
Lemma 3.2 short cofinal predicate	•	•	◦	•		•	•	◦	
Lemma 3.4 transfer (general form)	◦	◦	◦	•		•	◦	◦	
Lemma 3.5 $\gamma \geq \lambda$						•			
Lemmas 4.1–4.2 persistent covers, exact hulls				•		•			
Lemma 4.3 trace reconstruction	•	•	•	•		•	•	•	
Lemma 4.4 singular endpoint						•			
Theorem 5.1(2),(3) barrier	•	•	•	•		•	•	•	
Theorem 5.1(1) absorption				•		•			
Theorem 5.1(4) cover gap		•		•		•			
Corollary 5.3 sequences/embeddings escape			◦					•	
Corollary 5.4 failed-stationarity certificate	•								
Corollary 5.5 cover/approximation failure	•	◦	•	◦		◦		•	
Theorem 6.1 separation below λ	•	•	•	•		•	•	•	
item (1) no small cover / (2) no refinement / (3)		(1)		(3)		(2)			
ultrafilter									
Corollary 6.2 conditional inconsistency	•	•	•	•		•	•	•	
Corollary 6.3 extendible recovery	•	•	•	•		•	•	•	
Theorem 6.4 first regularization						•			
Lemma 7.1 (λ -cc form)	◦	◦	•	•		•	◦	•	
Theorem 7.2 proper ground	•	•	•	•		•		•	
λ strongly inaccessible in the ground								•	
Corollary 7.3 Ground Axiom obstruction	•	•	•	•		•		•	
Corollary 7.4 general / small-ground form	•		•				•		
Corollary 7.5 two strongly compacts	•					•		•	
Theorem 8.1, Corollary 8.2 projection width									•
Proposition 8.3 fresh cofinal set		◦	◦					◦	•
Theorem 9.4 finite selector families					•		◦		
Theorem 9.6 Icarus enrichment					•				
Theorem 9.8 finite labels							•		
Theorem 9.10 finite-valued transversal							•		

Result	R1	R2	R3	R4	R5	R6	R7	R8	R9
Theorem 10.1 $I3_{wf}(0) \Rightarrow CEx$									•
Corollary 10.6 calibration	•					◦			◦
Theorem 10.7 I0 package		•	•		•		•	•	
Theorem 11.2 internal unbounded range			•						
HKP reconstruction audited as known		•	◦			•	•		

Second round

Result	R10	R11	R12	R13	R14	R15	R16	R17	R18
Lemma 8.4(a) fixed small sets of ordinals	•	•	•	•	•	•	•	•	•
Lemma 8.4(b) for arbitrary short cofinal sets	•	•							•
for families in D_λ	•	•	•	◦	•	•	•	•	•
Theorem 8.5 (exacting)	•				◦			◦	•
minimal-rank-parameter device									•
Theorem 8.6 bounded separation	•				•			•	
Theorem 8.7, Cor. 12.11 laws and kernels	•								
Theorem 7.7, Cor. 7.8, 7.9					•				
Lemma 12.1 roots and orbits			•			•			
offsets $\kappa_n + t$				•	•			•	
critical-sequence class only	•	•					•		•
Theorem 12.3 rigidity from q as parameter		◦							•
Cor. 12.4(1) for all OD_{V_λ} sets at once								•	•
Cor. 12.4(2) small families, partial sections					◦	•		◦	•
Cor. 12.4(4) the model N_q		•							◦
Cor. 12.5(2) no thin complete section	•	•	•	•	•	•	•	•	•
λ full fibres			•	•	•	•		•	
small families of thin sections	•		◦	◦	•	•	◦	•	•
Remark 12.6(b) parameter-rank cutoff								•	
Remark 12.6(d),(e) recoverable classes, recodings		•							
Theorem 12.7, Cor. 12.8 width and phase			•						
Cor. 12.9(a) colourings						•	•	•	
Cor. 12.9(b) coarser relations					•		•		
Cor. 12.9(c) homomorphisms									•
Theorem 12.10 cofinal-output rules	•	•					•		•
Lemma 12.12, Theorem 12.13, Cor. 12.14				•					
Theorem 12.15(a) $V_{\lambda+1}$ with predicate				•			•		
Theorem 12.15(b),(c) inner models, Icarus	•	•	•	◦		•	•	•	•
Proposition 12.16 periodic classes; $ Q_\lambda = 2^\lambda$						•			
Theorem 10.7(6) I0 package	•	•	•	•	•	•	•	•	•

B Directory map

New	Original location under docs/research-reports/ (folder, or archive : folder)
01/	Large_Cardinals_Continuation (1)/Large_Cardinals_Continuation/
02/	Large_Cardinals_Research_Followup/
03/	Large_Cardinals_Research_Continuation/
04/	Large_Cardinals_Research_Continuation(1) (1)/Large_Cardinals_Research_Continuation/
05/	Large_Cardinals_Research_Continuation (1)/Large_Cardinals_Research_Continuation/
06/	Large_Cardinals_Research_Continuation (2)/
07/	Large_Cardinals_Continuation/
08/	Large_Cardinals_Continuation(1)/Large_Cardinals_Continuation/
09/	Large_Cardinals_Research_Continuation(1)/Large_Cardinals_Research_Continuation/
10/	Cardinals_Research_Continuation.zip : Cardinals_Continuation/
11/	Cardinals3_Research_Continuation.zip : Cardinals3/
12/	Large_Cardinals_Maximal_Width.zip : Large_Cardinals_Maximal_Width/
13/	Cardinals3_Continuation.zip : Cardinals3_Continuation/
14/	Cardinals_Continuation.zip : Cardinals_Continuation/
15/	Large_Cardinals_Orbit_Saturation.zip : Orbit_Saturation_Report/
16/	Cardinals_Continuation_Small_Fibres.zip : Cardinals_Continuation_Small_Fibres/
17/	Large_Cardinals_Thin_Sections.zip : Large_Cardinals_Thin_Sections/
18/	Cardinals3_Cofinal_Orbit_Rigidity.zip : Cardinals3_Continuation/

References

- [1] *Large cardinals at the inconsistency frontier: a unified assessment of ZFC-compatible hypotheses, their obstructions, and the evidence on both sides*. Research plan, 18 September 2026; docs/research-plan/Large_Cardinals_Unified_Report.tex.
- [2] G. Goldberg, reporting joint work with D. Blue. *Consistency beyond the Kunen inconsistency*. Lecture notes, Warsaw Inner Model Theory Meeting, 1 July 2026. <https://math.berkeley.edu/~goldberg/Slides/CoverExact.pdf>. Definitions 2.6, 3.1–3.2; Remark 2.4; Lemma 3.3; Proposition 3.4; Theorems 3.5–3.8; Problem 5.2.
- [3] G. Goldberg. The uniqueness of elementary embeddings. *J. Symbolic Logic* **89**(4) (2024), 1430–1454. doi:10.1017/jsl.2024.60. Author version used for numbering; earlier preprint arXiv:2103.13961 (numbering shifted).
- [4] J. P. Aguilera, J. Bagaria, P. Lücke. *Large cardinals, structural reflection, and the HOD Conjecture*. arXiv:2411.11568v4, 2025.
- [5] J. P. Aguilera, J. Bagaria, G. Goldberg, P. Lücke. *Large cardinals beyond HOD*. arXiv:2509.10254v1, 2025.
- [6] A. Andretta, V. Dimonte. *The iterability hierarchy above I3*. arXiv:1712.03877v2.
- [7] K. Kunen. Elementary embeddings and infinitary combinatorics. *J. Symbolic Logic* **36**(3) (1971), 407–413. doi:10.2307/2269948.
- [8] J. D. Hamkins, G. Kirmayer, N. L. Perlmutter. Generalizations of the Kunen inconsistency. *Ann. Pure Appl. Logic* **163** (2012), 1872–1890. arXiv:1106.1951.
- [9] J. Reitz. The Ground Axiom. *J. Symbolic Logic* **72**(4) (2007), 1299–1317. doi:10.2178/jsl/1203350787.